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Device for applying a knot to a suture

Abstract

Methods and apparatuses are disclosed for closing a patent foramen ovale. Some of the disclosed apparatuses include an elongate body having a proximal end and a distal end, with first and second suture clasp arms adapted to hold end portions of a suture when in an extended position. A first suture catch mechanism is slidably housed in the elongate body and moves in a proximal-to-distal direction to engage the suture end held by the first suture clasp arm, and a second suture catch mechanism is slidably housed in the elongate body and moves in a distal-to-proximal direction to suture end held by the second suture clasp arm. The first suture clasp arm can be positioned around the septum primum to deliver a suture thereto, and the second suture clasp arm can be positioned around the septum secundum to deliver a suture thereto.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS
(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.
BACKGROUND OF THE INVENTIONS
Field of the Invention
(2) Embodiments of the present inventions relate to suturing devices and methods. Specifically, preferred embodiments of the present inventions relate to suturing devices and methods for suturing a patent foramen ovale.
Description of the Related Art
(3) Health practitioners frequently use sutures to close various openings such as cuts, punctures, and incisions in various places in the human body. Because of their importance and frequent use, several types of sutures and devices for their implantation and extraction have been developed. These devices include needles having various shapes and sizes as well as devices for inserting and removing staples. Generally, sutures are convenient to use and function properly to hold openings in biological tissue closed thereby aiding in blood clotting, healing, and prevention of scaring.

However, there are some circumstances under which it is not feasible to use conventional sutures and suturing methods to close an opening. Some of these circumstances occur with incisions in arterial walls, or other internal bodily tissues. Here, catheter based devices and procedures have been suggested to close such openings.
(4) For example, during development of a fetus in utero, blood is generally oxygenated by the mother's placenta, not the fetus' developing lungs. Most of the fetus' circulation is shunted away from the lungs through specialized vessels or foramens, such as the foramen ovale. The foramen ovale is a flaplike opening between the atrial septa primum and secundum which serves as a physiologic conduit for right to left shunting between the atria. Typically, once the pulmonary circulation is established after birth, left atrial pressure increases, resulting in the fusing of the septum primum and septum secundum and thus the closure of the foramen ovale. Occasionally, however, these foramen fail to close and create hemodynamic problems, which may ultimately prove fatal unless treated. A foramen ovale which does not seal is defined a patent foramen ovale, or PFO.
(5) To close such PFOs, open surgery may be performed to ligate and close the defect. Such procedures are obviously highly invasive and pose substantial morbidity and mortality risks. Alternatively, catheter-based procedures have been suggested which involve introducing expandable structures through the patent foramen ovale to attempt to secure the tissue surrounding the patent foramen ovale, thereby blocking and sealing the patent foramen ovale. However, these structures involve support structures which may fail during the life of the patient and/or become dislodged, thereby re-opening the patent foramen ovale and possibly releasing the structure within the patient's heart. Thus, it would be advantageous to provide a simple, closure device and procedure for sealing a patent foramen ovale.
SUMMARY OF THE INVENTIONS
(6) Embodiments of the present inventions address the above problems by providing a suturing device and method for suturing biological tissue, such as, for example, an organ or blood vessel. The device is particularly well suited to suture a patent foramen ovale.
(7) One embodiment relates to a suturing device comprising an elongate body and at least one arm, more preferably first and second arms. Each of said arms has a suture mounting portion which mounts an end portion of a suture. The arms are mounted on the elongate body such that said suture mounting portions are movable away from said body to a first position and towards said body to a second position. The suturing device further comprises at least one needle, and preferably first and second needles, each of said needles having a distal end. Each of said needles is mounted such that the distal end of the needle is movable from a position adjacent said elongate body to a position away from said body, and towards the suture mounting portion of one of the arms when in said first position, wherein the respective distal ends of the first and second needles engage respective end portions of said suture. The suturing apparatus further comprises an actuator which drives the needles.
(8) In one embodiment, a suturing apparatus comprises an elongate body having a proximal end and a distal end. A first suture clasp arm is adapted to hold an end portion of a suture, the first suture clasp arm being extendable from the body from a retracted position to an extended position. A second suture clasp arm is adapted to hold an end portion of a suture, the second suture clasp arm being extendable from the body from a retracted position to an extended position. A first suture catch mechanism is slidably housed in the elongate body, the first suture catch mechanism being moveable in a proximal to distal direction to engage a distal end of the first suture catch mechanism with the suture end held by the first suture clasp arm when the first suture clasp arm is in the extended position. A second suture catch mechanism is slidably housed in the elongate body, the second suture catch mechanism being moveable in a distal to proximal direction to engage a distal end of the first suture catch mechanism with the suture end held by the second suture clasp arm when the second suture clasp arm is in the extended position.

(9) In another embodiment, a suturing apparatus for suturing a patent foramen ovale comprises an elongate body having a proximal end and a distal end configured to be delivered percutaneously into the patent foramen ovale. At least a first suture clasp arm is adapted to hold a first suture end portion. The first suture clasp arm is extendable from said body from a retracted position to an extended position and configured to be placed around one of the septum primum and septum secundum of the patent foramen ovale. At least a first suture catch mechanism is slidably housed in said elongate body. The first suture catch mechanism is movable through one of the septum primum and septum secundum of the patent foramen ovale to engage a distal end of the first suture catch mechanism with the first suture end portion held by the first suture clasp arm when the first suture clasp arm is in the extended position.
(10) In another embodiment, a system for suturing a patent foramen ovale comprises a first suturing apparatus and a second suturing apparatus. The first suturing apparatus comprises a first elongate body having a proximal end and a distal end, a first suture clasp arm adapted to hold a first suture end portion, and a first suture catch mechanism slidably housed in said elongate body. The first suture clasp arm is extendable from said body from a retracted position to an extended position and configured to be placed around the septum primum of the patent foramen ovale. The first suture catch mechanism is moveable in a proximal to distal direction through the septum primum of the patent foramen ovale to engage a distal end of the first suture catch mechanism with the first suture end portion held by the first suture clasp arm when the first suture clasp arm is in the extended position. The second suturing apparatus comprises a second elongate body having a proximal end and a distal end, a second suture clasp arm adapted to hold a second suture end portion, and a second suture catch mechanism slidably housed in said elongate body. The second suture clasp arm is extendable from said body from a retracted position to an extended position and configured to be placed around the septum secundum of the patent foramen ovale. The second suture catch mechanism is moveable in a distal-to-proximal direction through the septum secundum of the patent foramen ovale to engage a distal end of the second suture catch mechanism with the second suture end portion held by the second suture clasp arm when the second suture clasp arm is in the extended position.
(11) In another embodiment, a suturing apparatus for suturing a patent foramen ovale comprises an elongate body having a proximal end and a distal end, a first suture clasp arm, a second suture clasp arm, a first suture catch mechanism, and a second suture catch mechanism. The first suture clasp arm is adapted to hold a first suture end portion. The first suture clasp arm is extendable from said body from a retracted position to an extended position and configured to be placed around the septum primum of the patent foramen ovale. The second suture clasp arm is adapted to hold a second suture end portion. The second suture clasp arm is extendable from said body from a retracted position to an extended position and configured to be placed around the septum secundum of the patent foramen ovale. The first suture catch mechanism is slidably housed in said elongate body. The first suture catch mechanism is moveable in a proximal-to-distal direction through the septum primum of the patent foramen ovale to engage a distal end of the first suture catch mechanism with the first suture end portion held by the first suture clasp arm when the first suture clasp arm is in the extended position. The second suture catch mechanism is slidably housed in said elongate body. The second suture catch mechanism is moveable in a distal-to-proximal direction through the septum secundum of the patent foramen ovale to engage a distal end of the first suture catch mechanism with the second suture end portion held by the second suture clasp arm when the second suture clasp arm is in the extended position.
(12) In another embodiment, a method of closing a patent foramen ovale having a septum primum and a septum secundum is provided. An elongate body is advanced into a tunnel of a patent foramen ovale. A first suture clasp arm is extended from the elongate body from a retracted position to an extended position, the first suture clasp arm holding an end portion of a suture. The first suture clasp arm is positioned around one of the septum primum and the septum secundum. A

first needle positioned in the elongate body is advanced outwardly from the body through tissue of one of the septum primum and septum secundum and into engagement with the suture end held in the first suture clasp arm. The first needle is retracted into the elongate body with the suture end carried by the first needle. A second suture clasp arm is extended from the elongate body from a retracted position to an extended position, the second suture clasp arm holding an end portion of a suture. The second suture clasp arm is positioned around the other of the septum primum and the septum secundum. A second needle positioned in the elongate body is advanced outwardly from the body through tissue of the other of the septum primum and septum secundum and into engagement with the suture end held in the second suture clasp arm. The second needle is retracted into the elongate body with the suture end carried by the second needle. The elongate body is withdrawn from the tunnel of the patent foramen ovale. The septum primum and the septum secundum are drawn closed.

(13) In another embodiment, a method of closing a patent foramen ovale having a septum primum and a septum secundum is provided. A first suture clasp arm is positioned around one of the septum primum and the septum secundum. The first suture clasp arm holds a first suture end portion. A first needle is advanced through tissue of one of the septum primum and septum secundum and into engagement with the first suture end portion held in the first suture clasp arm. The first needle is retracted through tissue of one of the septum primum and septum secundum with the first suture end portion carried by the first needle. A second suture clasp arm is positioned around the other of the septum primum and the septum secundum. The second suture clasp arm holds a second suture end portion. A second needle is advanced through tissue of the other of the septum primum and septum secundum and into engagement with the second suture end portion held in the second suture clasp arm. The second needle is retracted through tissue of the other of the septum primum and septum secundum with the second suture end portion. The septum primum and the septum secundum are drawn closed.

(14) In another embodiment, a method of closing a patent foramen ovale having a septum primum and a septum secundum is provided. A first elongate body is advanced into a tunnel of a patent foramen ovale. A first suture clasp arm is extended from the first elongate body from a retracted position to an extended position. The first suture clasp arm holds a first suture end portion. The first suture clasp arm is positioned around one of the septum primum and the septum secundum. A first needle positioned in the elongate body is advanced outwardly from the body through tissue of one of the septum primum and septum secundum and into engagement with the first suture end portion held in the first suture clasp arm. The first needle is retracted into the first elongate body with the first suture end portion carried by the first needle. The first elongate body is withdrawn from the tunnel of the patent foramen ovale. A second elongate body is advanced near the tunnel of a patent foramen ovale. A second suture clasp arm is extended from the second elongate body from a retracted position to an extended position. The second suture clasp arm holds a second suture end portion. The second suture clasp arm is positioned around the other of the septum primum and the septum secundum. A second needle positioned in the second elongate body is advanced outwardly from the body through tissue of the other of the septum primum and septum secundum and into engagement with the second suture end portion held in the second suture clasp arm. The second needle is retracted into the second elongate body with the second suture end portion carried by the second needle. The second elongate body is withdrawn from the tunnel of the patent foramen ovale. The septum primum and the septum secundum are drawn closed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a method of providing access to an exemplifying use environment, such as a

patent foramen ovale.
(2) FIG. **2**A illustrates a side view of one embodiment of a suturing device.
(3) FIG. **2**B illustrates a side view of the distal end of the suturing device of FIG. **2**A.
(4) FIG. **2**C illustrates a cross-sectional view of the elongate tubular member of an embodiment of the suturing device taken along the line **2**C-**2**C of FIG. **2**B.
(5) FIG. **2**D illustrates a cross-sectional view of the spreader assembly of an embodiment of the suturing device taken along the line **2**D-**2**D of FIG. **2**B.
(6) FIG. **2**E illustrates a cross-sectional view of the distal tip of an embodiment of the suturing device taken along the line **2**E-**2**E of FIG. **2**B.
(7) FIG. **3** illustrates a side view of an embodiment of the suturing device with the suture clasp arms deployed.
(8) FIG. **4**A illustrates a perspective view of the suture clasp arms.
(9) FIG. **4**B illustrates a top plan view of an embodiment of a suture clasp arm having one suture clasp.
(10) FIG. **4**C illustrates a top plan view of an alternative embodiment of a suture clasp arm having two suture clasps.
(11) FIG. **5** is a side view of an embodiment of the suturing device showing suture portions positioned in the suture clasp arms.
(12) FIG. **6**A illustrates a top plan view of an embodiment of a spreader assembly showing a needle guide.
(13) FIG. **6**B illustrates a cross-sectional view of an embodiment of a spreader assembly showing proximal and distal needle guides.
(14) FIG. **7**A is a side view of an embodiment of a suture catch mechanism for engaging the proximal suture clasp arm.
(15) FIG. **7**B is a side view of an embodiment of a suture catch mechanism for engaging the distal suture clasp arm.
(16) FIG. **8**A is a side view of an embodiment of the suturing device illustrating the proximal and distal suture catch mechanisms in a stored position.
(17) FIG. **8**B is a side view of an embodiment of the suturing device illustrating the proximal suture catch mechanism in a deployed position.
(18) FIG. **8**C is a side view of an embodiment of the suturing device illustrating the distal suture catch mechanism in a deployed position.
(19) FIG. **9**A is a perspective view of one embodiment of a handle of the suturing device.
(20) FIG. **9**B is a perspective view of the handle of FIG. **9**A, with a portion of the housing removed.
(21) FIG. **10**A is a schematic representation an embodiment of the suturing device deployed in a PFO.
(22) FIG. **10**B is a schematic representation as in FIG. **10**A with the proximal suture clasp arm positioned around the septum primum.
(23) FIG. **10**C is a schematic representation as in FIG. **10**B showing the proximal needle engaging the proximal suture clasp arm.
(24) FIG. **10**D is a schematic representation as in FIG. **10**C showing the proximal needle and suture portion retracted through the septum primum.
(25) FIG. **10**E is a schematic representation as in FIG. **10**D with the distal suture clasp arm positioned around the septum secundum.
(26) FIG. **10**F is a schematic representation as in FIG. **10**E showing the distal needle engaging the distal suture clasp arm.
(27) FIG. **10**G is a schematic representation as in FIG. **10**F following retraction of the distal needle and suture portion through the septum secundum.
(28) FIG. **10**H is a schematic representation as in FIG. **10**G showing the suture portions positioned

through the septum secundum and septum primum, and the suturing device being withdrawn.
(29) FIG. **10**I is a schematic representation as in FIG. **10**H showing the suture portions positioned through the septum secundum and septum primum following withdrawal of the suturing device.
(30) FIG. **10**J is a schematic representation of an alternative embodiment showing the suture portions positioned through the septum secundum and septum primum following withdrawal of the suturing device.
(31) FIG. **10**K is a schematic representation of an alternative embodiment showing the suture portions positioned through the septum secundum and septum primum following withdrawal of the suturing device.
(32) FIG. **10**L is a schematic representation of a patch being delivered to the PFO.
(33) FIG. **11**A is a schematic representation of an alternative embodiment showing a suturing device with the distal clasp arm positioned around the septum secundum.
(34) FIG. **11**B is a schematic representation showing the suturing device of FIG. **11**A with the distal needle engaging the distal suture clasp arm.
(35) FIG. **11**C is a schematic representation showing the suturing device of FIG. **11**A with the proximal clasp arm positioned around the septum primum.
(36) FIG. **11**D is a schematic representation showing the suturing device of FIG. **11**A with the proximal needle engaging the proximal suture clasp arm.
(37) FIG. **11**E is a schematic representation showing the suture portions deployed following the steps of FIGS. **11**A-**11**D.
(38) FIG. **12**A is a schematic representation of an alternative embodiment of a suturing device being delivered through a tunnel of the PFO.
(39) FIG. **12**B is a schematic representation of the suturing device of FIG. **12**A showing a distal suture clasp arm engaging the septum secundum.
(40) FIG. **12**C is a schematic representation of the suturing device of FIG. **12**A showing the distal needle engaging the distal suture clasp arm.
(41) FIG. **12**D is a schematic representation of the suturing device of FIG. **12**A showing the distal needle retracted from the septum secundum.
(42) FIG. **12**E is a schematic representation of the suturing device of FIG. **12**A showing the suturing device partially withdrawn from the tunnel of the PFO and the suture clasp arms retracted into the device.
(43) FIG. **12**F is a schematic representation of the suturing device of FIG. **12**A showing the suturing device advanced further into the left atrium with the proximal and distal suture clasp arms extended from the suturing device.
(44) FIG. **12**G is a schematic representation of the suturing device of FIG. **12**A showing a proximal suture clasp arm engaging the septum primum.
(45) FIG. **12**H is a schematic representation of the suturing device of FIG. **12**A showing the proximal needle engaging the proximal suture clasp arm.
(46) FIG. **12**I is a schematic representation of the suturing device of FIG. **12**A showing the proximal needle retracted from the septum primum.
(47) FIG. **12**J is a schematic representation of the suturing device of FIG. **12**A showing the suturing device advanced further into the left atrium.
(48) FIG. **12**K is a schematic representation of the suturing device of FIG. **12**A showing the suture clasp arms retracted into the suturing device.
(49) FIG. **12**L is a schematic representation of the suturing device of FIG. **12**A showing the suturing device being withdrawn from the PFO.
(50) FIG. **13** illustrates a side view of one embodiment of a suturing device.
(51) FIG. **14**A illustrates a side view of the distal end of the suturing device of FIG. **13**.
(52) FIG. **14**B illustrates a side view of a distal end of a suturing device of one embodiment.
(53) FIG. **15**A illustrates a cross-sectional view of the elongate tubular member of an embodiment

of the suturing device of FIGS. **13** and **14**A, taken along the line **15**A-**15**A of FIG. **14**A.
(54) FIG. **15**B illustrates a cross-sectional view of the elongate tubular member of an embodiment of the suturing device of FIG. **14**B, taken along the line **15**B-**15**B of FIG. **14**B.
(55) FIG. **16** illustrates a perspective view of a spreader assembly of an embodiment of the suturing device.
(56) FIG. **17** illustrates another perspective view of the spreader assembly of FIG. **16**.
(57) FIG. **18**A illustrates a cross-sectional view of a distal end of an elongate tubular member, the spreader assembly of FIGS. **16** and **17**, and a housing taken along the line **18-18** of FIG. **15**A.
(58) FIG. **18**B illustrates a cross-sectional view of a distal end of an elongate tubular member, a spreader assembly, and a housing showing features for limiting the range of travel of suture catch mechanisms.
(59) FIG. **19** illustrates a side view of an embodiment of the suturing device with the suture clasp arms deployed and various internal features shown in phantom.
(60) FIG. **20** illustrates a perspective view of a suture clasp arm.
(61) FIG. **21** illustrates a perspective view of a suture clasp arm.
(62) FIG. **22** illustrates a cross-sectional view of a suture clasp arm and a needle.
(63) FIG. **23** illustrates a perspective view of the suture clasp arm and the needle of FIG. **22**.
(64) FIG. **24** illustrates a perspective view of a housing of an embodiment of the suturing device.
(65) FIG. **25** illustrates another perspective view of the housing of FIG. **24**.
(66) FIG. **26** illustrates a top view of a distal end of an embodiment of a suturing device with various features shown in phantom lines.
(67) FIG. **27** is a perspective view of one embodiment of a handle of the suturing device.
(68) FIG. **28**A is a cross-sectional view of the handle of FIG. **27** taken along the line **28**A-**28**A shown in FIG. **27**.
(69) FIG. **28**B is a cross-sectional view of the handle of FIG. **27** taken along the line **28**B-**28**B shown in FIG. **27**.
(70) FIG. **29**A is a schematic representation an embodiment of the suturing device deployed in a PFO.
(71) FIG. **29**B is a schematic representation as in FIG. **29**A with a distal suture clasp arm positioned around the septum primum.
(72) FIG. **29**C is a schematic representation as in FIG. **29**B showing the proximal needle engaging the distal suture clasp arm.
(73) FIG. **29**D is a schematic representation as in FIG. **29**C showing the proximal needle and suture portion retracted through the septum primum.
(74) FIG. **29**E is a schematic representation as in FIG. **29**D showing the suturing device positioned to permit a proximal suture clasp arm to extend from the suturing device.
(75) FIG. **29**F is a schematic representation as in FIG. **29**E with the proximal suture clasp arm positioned around the septum secundum.
(76) FIG. **29**G is a schematic representation as in FIG. **29**F showing the distal needle engaging the proximal suture clasp arm.
(77) FIG. **29**H is a schematic representation as in FIG. **29**G following retraction of the distal needle and suture portion through the septum secundum.
(78) FIG. **29**I is a schematic representation as in FIG. **29**H showing the suture portions positioned through the septum secundum and septum primum, and the suturing device being withdrawn.
(79) FIG. **29**J is a schematic representation as in FIG. **29**D showing the suturing device of FIGS. **14**B and **15**B positioned to permit the proximal suture clasp arm to extend from the suturing device and a second guide wire extended.
(80) FIG. **29**K is a schematic representation as in FIG. **29**J with the proximal suture clasp arm positioned around the septum secundum.
(81) FIG. **29**L is a schematic representation as in FIG. **29**K showing the distal needle engaging the

proximal suture clasp arm.
(82) FIG. **29**M is a schematic representation as in FIG. **29**L following retraction of the distal needle and suture portion through the septum secundum.
(83) FIG. **30** is a perspective view of one embodiment of a handle of a knot placement device.
(84) FIG. **31** is a cross-sectional view of the handle of FIG. **30** taken along the line **31-31** shown in FIG. **27**.
(85) FIG. **32** is a cross-sectional view of a distal end of a knot placement device.
(86) FIG. **33** is a perspective view of one embodiment of a handle of a suturing device.
(87) FIG. **34**A is a cross-sectional view of the handle of FIG. **33** taken along the line **34**A-**34**A shown in FIG. **33**.
(88) FIG. **34**B is a cross-sectional view of the handle of FIG. **33** taken along the line **34**B-**34**B shown in FIG. **33**.
(89) FIG. **35** is a side view of an actuator and a follower of the handle of FIG. **33**.
(90) FIG. **36** is a side view of one embodiment of a system of suturing devices.
(91) FIG. **37** is a perspective view of a distal end of a first suturing device of the system of FIG. **36**.
(92) FIG. **38** is a perspective view of a distal end of a second suturing device of the system of FIG. **36**.
(93) FIG. **39**A is a schematic representation an embodiment of the first suturing device of the FIG. **37** deployed in a PFO.
(94) FIG. **39**B is a schematic representation as in FIG. **39**A with a suture clasp arm positioned around the septum primum.
(95) FIG. **39**C is a schematic representation as in FIG. **39**B showing a needle engaging the suture clasp arm.
(96) FIG. **39**D is a schematic representation as in FIG. **39**C showing the needle and suture portion retracted through the septum primum.
(97) FIG. **39**E is a schematic representation as in FIG. **39**D showing the second suturing device of FIG. **38** positioned to permit a suture clasp arm to extend from the second suturing device and a second guidewire extended.
(98) FIG. **39**F is a schematic representation as in FIG. **39**E with the suture clasp arm positioned around the septum secundum.
(99) FIG. **39**G is a schematic representation as in FIG. **39**F showing a needle engaging the suture clasp arm.
(100) FIG. **39**H is a schematic representation as in FIG. **39**G following retraction of the needle and suture portion through the septum secundum.
(101) FIG. **39**I is a schematic representation as in FIG. **39**H showing the suture portions positioned through the septum secundum and septum primum, and the second suturing device being withdrawn.
(102) FIG. **39**J is a schematic representation as in FIG. **39**I showing the suture portions being joined by a first knot following withdrawal of the suturing device.
(103) FIG. **39**K is a schematic representation as in FIG. **39**J showing the first knot being positioned between the septum secundum and septum primum.
(104) FIG. **40**A is a schematic representation as in FIG. **39**D showing the second suturing device positioned to permit the suture clasp arm to extend from the second suturing device and the second guide wire extended through the PFO.
(105) FIG. **40**B is a schematic representation as in FIG. **40**A with the suture clasp arm positioned in the PFO around the septum secundum.
(106) FIG. **40**C is a schematic representation as in FIG. **40**B showing the needle engaging the suture clasp arm.
(107) FIG. **40**D is a schematic representation as in FIG. **40**C following retraction of the needle and suture portion through the septum secundum.

(108) FIG. **40**E is a schematic representation as in FIG. **40**D showing the suture portions positioned through the septum secundum and septum primum, and the second suturing device being withdrawn

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(109) Embodiments of the present invention provide suturing devices and methods for closing an opening in biological tissue, a body lumen, hollow organ or other body cavity. The suturing devices and their methods use are useful in a variety of procedures, such as treating (closing) wounds and naturally or surgically created apertures or passageways. For example, the suturing devices may be used to seal an opening in the heart wall such as an atrial septal defect, a patent ductus arteriosus or a patent foramen ovale. In addition, the suturing devices may be used to close or reduce a variety of other tissue openings, lumens, hollow organs or natural or surgically created passageways in the body. Also, the suturing devices may be used to suture prosthetics, synthetic materials, or implantable devices in the body. For example, the devices may be used to suture pledget within the body.

(110) FIG. **1** illustrates one embodiment in an exemplifying use environment for closing a patent foramen ovale (PFO). Adaption of the devices and methods disclosed herein for closing a PFO may also be made with respect to procedures for closing other bodily tissue openings, lumens, hollow organs or natural or surgically created passageways and procedures for suturing prosthetics, synthetic materials, or implantable devices in the body. As depicted by FIG. **1**, a guidewire **10** is advanced into the right atrium **2** of the heart **9** through the inferior vena cava **3**. It is anticipated that the heart may be accessed through any of a variety of pathways, such as through the inferior vena cava **3** via a femoral access site, through the superior vena cava **5** via the subclavian or jugular veins, or any other venous or arterial access sites. The guidewire **10** can then be further positioned in the tunnel or opening of the patent foramen **8** ovale between the septum primum **7** and septum secundum **6**. With the guidewire **10** in place, the physician can insert a sheath **11** to the right atrium. This sheath **11** is typically a single lumen catheter with a valve on its proximal end. The valve is used to prevent extraneous bleed back or to introduce medication into the patient's body. The sheath **11** can be placed at or near the tunnel of a patent foramen ovale **8**. The suturing device **100**, described further below, can then be advanced to the PFO **8** through the lumen of the sheath **11**. In an alternative embodiment, the suturing device **100** can be advanced over the guidewire **10** and positioned in the opening of the patent foramen ovale **8** without the need to insert an introducer **11**. Other methods of accessing the PFO or other bodily locations.

(111) FIG. **2**A shows one embodiment of the suturing device **100** for suturing an opening in a vessel wall and other biological tissue. While the device will be described in reference to suturing an opening in the heart wall such as a patent foramen ovale (PFO), the device could be used to close other openings in the heart wall, such as a patent ductus arteriosus (PDA) or an atrial septal defect (ASD), other puncture wounds in bodily tissue, or the like, or to perform other procedures as described above. The suturing device **100** comprises an elongated tubular member **20** having a spreader assembly **30** (shown in FIG. **2**B) connected to the distal end of the elongated tubular member **20** for positioning in the opening of the PFO. A handle **200** is provided at the proximal end of the tubular member **20**. The axial length and flexibility of the elongated tubular member **20** is sufficient to percutaneously access the patient's vasculature and advance the elongate tubular member **20** through the venous system to the patient's heart with the proximal end of the device remaining outside the patient's body. However, the length of the device may vary depending upon the intended access point and pathway to the heart. For example, for a femoral access point and pathway via the inferior vena cava, the axial length of the elongate tubular member **20** can be between about 70-120 cm, alternatively between about 80-100 cm, alternatively about 90 cm.

(112) As shown in more detail in FIG. **2**C, the elongate tubular member **20** has a plurality of lumens extending along the axial length. The multi-lumen elongate tubular member **20** can be manufactured in accordance with any of a variety of techniques known to those skilled in the art.

For example, in some embodiments, the elongate tubular member **20** can be formed from a multi-lumen extruded plastic tubing, such as a polyester, polyethelyne, polymide, nylon or any other suitable material known to those skilled in the art. The elongate tubular member **20** comprises four central lumens **21**, **22**, **23** and **24** vertically stacked along a central axis of the elongate tubular member **20**. In some embodiments, the central lumens may be surrounded by two semicircular or D-shaped lumens **25** and **26** extending axially on opposite sides of the elongate tubular member **20**. In some embodiments, as will be discussed below in more detail, the central lumens **21**, **22**, **23**, and **24** will be used to provide access through the elongate tubular member **20** for a guidewire, to provide access for an actuating rod connected to the suture clasp arms, and to house one or more suture catch mechanisms or needles. The semi-circular shaped lumens **25** and **26** can be used to deliver one or more sutures to the distal end of the elongate tubular member **20**.
(113) The spreader assembly **30** is bonded, or otherwise joined, to the distal end of the elongated tubular member **20**, for example with epoxy or any other suitable technique known to those skilled in the art. Alternatively, the spreader assembly may be integral with the elongate tubular member **20** As shown in FIG. **2**D, the spreader assembly comprises central lumens **131**, **132**, **133** and **134** vertically stacked along a central longitudinal axis of the spreader assembly **30**. When the spreader **30** and the elongate tubular member are properly connected, the central lumens **21**, **22**, **23** and **24** of the elongate tubular member **20** and the central lumens **131**, **132**, **133** and **134** of the spreader assembly **30** are preferably substantially aligned to provide continuous passageways through the elongate tubular member **20** and spreader assembly **30**. A metal casing, or bullet, **40**, having a length greater than the length of the spreader **30** and an inner diameter substantially the same as the outer diameter of the spreader assembly **30** and elongate tubular member **20** is placed over the connection between the elongate tubular member **20** and the spreader assembly **30** to maintain the proper alignment of the internal lumens of the elongate tubular member **20** and spreader assembly **30**. The bullet comprises openings **41**A and **41**B located on the opposite sidewalls of the bullet **40** for allowing the release and deployment of suture clasp arms housed within the spreader assembly **30**. Accordingly, the openings **41**A and **41**B are sized and shaped to permit the suture clasp arms to fully extend from the spreader assembly **30**.
(114) In some embodiments, the bullet **40** has a length such that when the proximal end of the bullet is positioned over the connection between the spreader assembly **30** and the elongate tubular member **20**, its distal end extends beyond the distal end of the spreader assembly **30**. A distal tip **70** which may be rounded or atraumatic can be bonded or adhered to the distal end of the spreader assembly **30**, for example with epoxy or any other suitable technique known to those skilled in the art. As shown in FIG. **2**E, the distal tip can have at least one central lumen **172** that is axially aligned with a central lumen **132** of the spreader **30** and a central lumen **22** of elongate tubular member **20** for providing a continuous passageway through the entire length of suturing device **100**, for example for a guidewire. In addition, the distal tip **70** may have one or more additional lumens **173** that can be aligned with lumens in the spreader assembly **30** and elongate tubular member **20** to provide additional continuous passageways, for example for housing a suture catch mechanism. Here, as discussed above, the outer diameter of the distal tip is substantially the same size as the inner diameter of the bullet **40** such that when the distal end of the bullet **40** is positioned over the connection between the distal tip **70** and the spreader assembly **30**, the bullet maintains the proper alignment of the internal lumens of the distal tip **70** and spreader assembly **30**.
(115) As shown in FIG. **2**B, a pair of suture clasp arms **31**A and B are housed in recesses **41**A and **41**B in the central portion of the spreader assembly **30**. During storage and delivery of the suturing device, the suture clasp arms **31**A and **31**B are situated parallel to the longitudinal axis of the suturing device such that the outer walls of the suture clasp arms do not extend beyond the outer diameter of the spreader assembly **30**. After insertion and positioning of the suturing device at the PFO or other opening, the arms **31**A and B can be deployed to the position shown in FIG. **3**. To deploy the suture clasp arms, the suture clasp arms **31**A and B are connected to an actuating rod **35**

which extends through the passageway formed by lumen **134** in the spreader assembly and central lumen **24** in the elongate tubular member **20**. For example, in certain embodiments, the proximal suture clasp arm **31**A and the distal suture clasp arm **31**B may be manufactured as a single component, or alternately fixedly connected at a central connection point. Here, the distal end of the actuating arm **35** may be connected to the distal suture clasp arm **31**B, offset from the middle, or central connection point, of the suture clasp arms such that proximal retraction of the actuating arm **35** will pull on the distal suture clasp arm **31**B creating a counterclockwise torquing force on the suture clasp arms which will cause both suture clasp arms **31**A and B to flip out from the spreader assembly **30**.

(116) When deployed, the suture clasp arms **31**A and B extend from the suturing device **100** in opposite directions along the longitudinal axis of the device. Preferably, the arms **31**A and **31**B form an acute angle with the longitudinal axis of the spreader. In some embodiments, a first arm **31**A extends outward toward the proximal end of the spreader assembly **30** at an angle of between about 35-55.degree., alternatively about 40-50.degree., alternatively about 45.degree. with respect to the longitudinal axis of the spreader assembly **30**, while the second arm **31**B extends outward toward the distal end of the spreader assembly **30** at the same angle as the first arm **31**A with respect to the longitudinal axis of the spreader assembly **30**.

(117) As shown in FIG. **3**, the suture clasp arms **31**A and **31**B may be deployed simultaneously to extend equally in opposite directions with respect to the suturing device. In some embodiments, the suture clasp arms **31**A and B can be independently actuated to be individually deployed depending upon the location of the tissue portion to be sutured. However, in some embodiments, it may be advantageous to simultaneously deploy both suture clasp arms **31**A and B, even though only one of the suture clasp arms will be engaged at that location. For example in closing a PFO, the suturing device will first be positioned with respect to a first tissue portion to be sutured and then moved to a second position for suturing a second tissue portion. However, the second suture clasp arm **31**B can still be deployed at the first location to provide mechanical support and stabilization for positioning the first suture clasp arm **31**A proximal to the first tissue portion. For example, as shown in FIG. **10**B, when closing a PFO, the suturing device **100** is first positioned such that suture clasp arm **31**A extends around the septum primum **7** such that suture portion **52**A may be engaged and pulled though the septum primum **7** to draw it toward the septum secundum **6**. Here, the non-engaged suture clasp arm **31**B is still extended toward and contacts the septum secundum **6** which pushes the suturing device **100** toward the septum primum **7** thereby assisting in placement of the suture clasp arm **31**A around the septum primum **7** and stabilizing the suturing device **100** during deployment of the suture catch mechanism to engage the suture.

(118) As shown in FIG. **5**, each of the suture clasp arms **31**A, **31**B has a suture clasp **33** for receiving and holding a suture **50**. The suture clasp **33** can be a circular opening with a diameter sized to securely receive and hold a loop of suture **50**. For example, as described in U.S. Pat. No. 6,562,052, the entirety of which is hereby incorporated by reference, the suture can comprise a length of suture **50** having a loop formed at each end of the suture. Other details regarding the apparatuses and methods of suturing devices that may be utilized with the embodiments disclosed throughout this specification are also found in U.S. Pat. No. 6,562,052 and are hereby incorporated by reference. The diameter of the suture loops and the inner diameter of the suture clasp **33** are preferably substantially the same such that the suture loops can be securely positioned in the suture clasp **33** during deployment of the suture clasp arms. The suture clasp **33** is advantageously angled such that when the suture clasp arm is deployed at an angle, the suture clasp **33** will hold the suture loop perpendicular to the path of the suture catch mechanism. In some embodiments, the suture clasp arm **31** can have a tab or slot **34** located on the distal end of the suture clasp arm **31** for guiding the suture loop into the suture catch **33** at the proper angle.

(119) In use, as shown in FIG. **5**, the suture **50** can be housed in one of the outer D-shaped, or suture, lumens **25**, **26** of the elongate tubular member **20**. The suture lumen **26** has a port or

opening **42** on the distal end of the elongate tubular member **20**. The suture **50** can be advanced through the opening **42** such that the suture ends, or end portions **52**A and **52**B, extend outside of the suturing device **100**. One of the suture end portions **52**A may extend from the opening **42** to the suture clasp arm **31**A. The other suture end portion **52**B may extend along the length of the bullet and into a side opening **71** in the distal tip **70**. The suture portion **52**B may extend out of a distal opening **73** in the distal tip, and then loop back into the opening **73**. The suture end portion **52**B then extends through side opening **71** to the suture clasp arm **31**B.
(120) The suture ends are positioned in the suture clasps **33**A and **33**B of the suture clasp arms **31**A and **31**B and held securely there until they are engaged and removed by a suture catch mechanism. As discussed above, the suture clasp arms **31**A and **31**B comprise slots **34**A and **34**B to guide the suture into the suture clasps **33** at the proper angle and assist in maintaining the suture loops in the suture clasp until they are engaged by the suture catch mechanisms. When the suture catch mechanism engages the loops of the suture ends, the suture loops will slide out of the suture clasp **33** and be released by the suture clasp arm **31**.
(121) In some embodiments, the suture clasp arms **31**A and **31**B may further comprise additional suture clasp(s) for holding additional suture portions. For example, in an alternative embodiment, shown in FIG. **4**C, each suture clasp arm can be configured to deploy two sutures from the suturing device to a particular location for suturing a single tissue portion. Here, each suture clasp arm **431** has two suture clasps **433**A and **433**B for receiving two different suture portions **150** and **151**. In addition, in some embodiments, each suture clasp arm can further include two slots **434**A and **434**B for guiding the suture portions **150** and **151** into the suture clasps **433**A and **433**B.
(122) As shown in more detail in FIGS. **6**A-B, the spreader assembly **30** includes a plurality of needle guides **60**A and **60**B for guiding a plurality of suture catch mechanisms, such as a needle or other penetrating mechanism, towards the deployed suture clasp(s). In one embodiment, the spreader assembly **30** may comprise two needle guides **60**A and **60**B located on the distal and proximal ends of the spreader assembly for guiding two needles toward proximally and distally deployed suture arms **31**A and **31**B depicted in FIG. **3**. Each of the needle guides **60**A and **60**B comprises an angled groove or channel in the sidewall of the spreader assembly **30** such that it will deflect a suture catch mechanism, or needle, exiting the suture assembly **30** along a path that intercepts the suture clasps **33**A and **33**B of the suture clasp arms when the suture clasp arms **31**A and **31**B are in a deployed position. For example, in some embodiments, the suture arms **31**A and **31**B may be sized such that a groove having an angle of between about 10-35.degree., alternatively about 15-25.degree., alternatively about 19.degree. with respect to the longitudinal axis of the spreader assembly will spread the needles to the proper angle to engage the suture loops in the deployed suture clasps **33**. In use, the proximal needle guide **60**A is aligned with lumen **131** of the spreader assembly and lumen **21** of the elongate tubular member such that when a suture catch mechanism is advanced through the passageway formed by lumens **131** and **21**, the distal end of the needle will be deflected by the needle guide **60**A towards the deployed suture clasp arm **31**A. Likewise, distal needle guide **60**B is aligned with lumen **173** of the distal tip such that when the distal tip of a needle housed in lumen **173** is extended towards suture clasp arm **31**B, the needle will be deflected toward the suture clasp **33**.
(123) In some embodiments, as depicted in FIG. **7**A, the suture catch mechanism **161** which is configured for engaging the suture clasp arm **31**A extending toward the proximal end of the spreader assembly **30** comprises an elongate, straight needle **161**. As shown in FIG. **7**A, the needle **161** has a sharp, distal tip **163** for penetrating a tissue portion positioned between the suturing device and the deployed suture clasp arm **31**A and a notch or groove **162** on the distal tip **162** for engaging the loop of suture portion **52**A held by suture clasp **33**. The notch or groove **162** is shaped or angled upward in order to grasp and dislodge the loop from the suture clasp **33** and retain the suture portion **52**A against the groove **162** as the needle is pulled back through the suture clasp **33** and tissue portion and retracted back into the suturing device **100**.

(124) As shown in FIG. **7**B, suture catch mechanism **165**, which is configured to engage a suture clasp arm **31**B extending towards the distal end of the spreader assembly **30** comprises an elongate needle **165** having a distal portion **166** bent approximately 180 degrees such that a portion of the needle is turned back upon itself. The turned portion **166** of the needle **165** has a length sufficient to extend from the spreader assembly **30** and engage the suture clasp **33**B on suture clasp arm **31**B when suture clasp arm **31**B is in a fully deployed position. Similar to elongate, straight needle **161**, the turned back portion **166** has a sharp distal tip **167** for penetrating a tissue portion positioned between the suturing device and the deployed suture clasp arm **31**B and a notch or groove **168** on the distal tip **167** for engaging a loop of suture help by suture clasp arm **31**B. The notch or groove **168** on the needle is configured to grasp and retain the loop of suture portion **52**B from the suture catch **33**B as the turned portion **166** of the needle **165** (FIG. **8**B) is conveyed through the tissue portion and retracted back into the groove **173** (FIGS. **8**B and **8**C) of the distal tip **70** of the suturing device **10**.
(125) The needles **161** and **165** are flexible and are preferably made of a material with shape memory, such as nickel titanium or NITINOL. Alternatively, the needles **161** and **165** may be comprised of spring steel, surgical stainless steel or any variation thereof.
(126) In use, as shown in FIG. **8**A, the needles **161** and **165** can be housed within a central passageway formed by lumens within the elongate tubular member, spreader assembly and distal tip. Here, suture catch mechanism **161** is slidably housed in the passageway formed by the central lumens **21** of the elongate tubular member **20** and central lumen **131** of the spreader assembly while suture catch mechanism **165** is slidably housed in the passageway formed by central lumen **23** of the elongate tubular member, central lumen **133** of the spreader assembly and lumen **173** of the distal tip **70**.
(127) As shown in FIG. **8**B, for suture arm **31**A extending outward toward the proximal end of the spreader assembly **30**, the proximal end of the needle guide **60**A is aligned with a lumen **131** (see FIG. **2**B) extending though the spreader assembly **30** such that when needle **161**, or other suture catch mechanism, is advanced through the passageway formed by central lumen **21** in the elongate tubular member **20** and central lumen **131** in the spreader assembly **30**, the distal end of the needle will exit the suturing device through central lumen **131** and be advanced along the groove of the needle guide **60**A. The needle guide **60**A then deflects the distal end of needle **161** outward along the angle of the groove to penetrate suture clasp **33**A on the suture clasp arm **31**A and engage the suture portion **52**A held by the suture clasp **33**A. Once the needle **161** has engaged the suture portion **52**A, the needle **161** may be retracted back into central lumen **131** along with the suture portion **52**A held by groove **162** on the distal end of needle **161**. (See FIG. **8**C.)
(128) As shown in FIG. **8**C, for suture arm **31**B, extending outward toward the distal end of the spreader assembly in a deployed position, the distal end of needle guide **60**B is aligned with a lumen **173** in the distal tip of the suturing device. Here, before insertion, the needle **165** has been advanced through central lumen **123** in the elongate tubular member beyond the spreader assembly **30** and positioned a slot or cavity **173** within the distal tip **70** of the suturing device that has been aligned with the central lumen **123** of the elongate tubular member. The cavity **173** has a diameter sized to receive the distal end of the needle **165**, including the turned portion **166**. The turned portion **166** is positioned in cavity **173** such that the distal tip **167** is aligned with the needle guide **60**B located on the distal end of the spreader assembly **30**. Thus, in use, when the proximal end of the needle **165** is pulled back through the central lumen **123** of the elongate tubular member **20**, the turned portion **166** of the needle will be advanced along the groove of the needle guide **60**B and deflected outward along the angle path of the groove to penetrate the deployed suture clasp **33**B on the suture clasp arm **31**B and engage suture portion **52**B held therein. Once the needle **165** has engaged the suture portion **52**B, the proximal end of the needle **165** may then be pushed forward through the central lumen **123** of the elongate tubular member which will cause the bent portion **166** of the needle to be retracted along needle guide **60**B into cavity **173** along with the suture

portion **52**B held by the groove **168** on the distal end of the needle **165**.
(129) In some embodiments, as discussed above, the suture catch assembly can comprise two suture catch mechanisms, or needles housed on opposite sides of the elongate tubular member **20** for engaging the two suture clasp arms. The needles may be independently actuated such that the needles may be independently deployed. In an alternative embodiment, the suture catch assembly may comprise four needles, for example two needles housed on each side of the elongate tubular member for engaging multiple suture portions in each of the suture clasp arms. Here, each of the needles may be independently actuated or alternatively both needles for engaging a single suture clasp arm may be jointly actuated such that they are deployed at the same time.
(130) Referring now to FIGS. **9**A-**9**B, the proximal end of the needles **161** and **165** extend through the central lumens **21** and **23** of the elongate tubular member and terminate at a connection to actuating levers **210** and **216** on the suturing device handle **200**. The handle **200** includes a housing **201** which is attached to the proximal end of the elongate tubular member **20**. The housing **201** has an aperture **202** providing a passageway between the handle **200** and the multiple lumens of the elongate tubular member **20**. One or more levers or buttons **210**, **216**, **240** and **250** may extend from the housing **201**. The levers or buttons **210**, **216**, **240** and **250** can be connected to actuating rods or mechanisms for deploying and/or retracting the suture clasp arms and suture catch mechanisms moveably housed in the suturing device.
(131) With reference to FIGS. **9**A-**9**B, in some embodiments, the handle **200** can have two actuating levers **210** and **216** for deploying and retracting the suture catch mechanisms such as needles **161** and **165** of the suturing device. For example, in one embodiment, the proximal end of needle **165** extends from the proximal end of central lumen **23** in the elongate tubular member through opening **202** into the handle **200** and is inserted into an opening **211** in a lever needle holder **212**. The needle **165** may be securely connected to opening **211** in the lever needle holder **212** by friction fit in the opening **211**, by bonding with epoxy, or any other suitable technique known to those in the art. The lever needle holder **212** is permanently connected to a needle deployment lever **210** extending from the handle housing **201**, such that in use, when the physician pushes the lever forward or pulls the lever back, the needle deployment lever **210** will advance or retract the needle holder **212** longitudinally along an axis of the handle **200** and thus advance or retract the attached needle **163** longitudinally within the lumen **23** of the elongate tubular member **20**. The housing **200** comprises a fixed length opening **213** surrounding the needle lever **210** which provides a limit to the distance the needle **163** can be advanced or retracted by limiting the distance over which the lever **210** can be pushed or pulled. In general, the axial length of the opening **213** should be sufficient to permit the turned distal end of the needle **163** to be extended from a pre-deployment position within the suturing device **100** to engage the distal suture clasp arm **31**A in a fully deployed position, as discussed above. However, limiting the distance the needle can be deployed may advantageously prevent the needle from being advanced too far and potentially puncturing adjacent tissue.
(132) The proximal end of needle **161** is likewise configured to extend through central lumen **21** of the elongate tubular member **20** and is inserted into an opening **214** in a lever needle holder **215**. The lever needle holder **215** is permanently connected to a second needle deployment lever **216** extending from the handle housing **201**, such that in use, when the physician pushes the lever forward or pulls the lever **216** back, the needle deployment lever **216** will advance or retract the needle holder **215** and thus advance or retract the attached needle **161**. The housing **201** comprises a fixed length opening **217** surrounding the needle lever **216** which provides a limit to the distance the needle **161** can be advanced or retracted by limiting the distance over which the lever **216** can be pushed or pulled. In general, the axial length of the opening **217** should be sufficient to permit the distal end of the needle **161** to be extended from a pre-deployment position within the elongate tubular member **20** to engage the proximal suture clasp arm **31**A in a fully deployed position, as discussed above.

(133) Deployment of the suture claps arms **31**A and **31**B can be performed by depressing a button **240** located on the handle **200**. The button **240** may be connected to an arm puller mechanism **241** which is configured to be extended and retracted along a longitudinal axis of the handle **200**, which is operably connected to actuating rod **35**. For example, in some embodiments, the arm puller mechanism may have a slot configured to receive the proximal end of the actuating rod **35**.
(134) The actuating rod **35** extends through central lumen **24** of the elongate tubular member **20** and is connected to the suture clasp arms **31** such that proximal retraction of the actuating rod **35** causes the suture clasp arms **31**A and B to extend from the spreader assembly **30**. For example, as discussed above, in some embodiments, the actuating rod **35** can be connected to the distal suture clasp arm **31**B at a location that causes the suture clasp arms **31**A and **31**B to swing out from the spreader assembly **30** in a counter-clockwise direction. In use, when button **240** is depressed, the arm pull mechanism **241** is pulled back against spring **251** causing the actuating rod **35** to be retracted though central lumen **24**, thus deploying the suture clasp arms. When button **240** has been completely depressed, the button **240** engages a lip **245** of the arm puller mechanism **241** and maintains the arm puller mechanism **241** in a locked, retracted position compressing spring **251**. A second button **250** is located on handle **200**, proximal to button **240**. Button **250** has a lever **252** extending within the housing **201** which is configured to engage an edge of button **240** and raise the button **240** from lip **245** of the arm puller mechanism. Once the arm puller mechanism **241** has been released, spring **251** may expand to an uncompressed state, thereby pushing the arm puller mechanism **241** forward. Forward motion of the arm puller mechanism advances the attached actuating rod **135** which transmits a clockwise rotational force to the suture clasp arms causing the suture clasp arms to be retracted within the spreader assembly **30**. Further details of actuation mechanisms as well as other devices and methods that may incorporated with the embodiments described throughout this specification are described in U.S. Patent Publication No. 2006-0069397 A1, published Mar. 30, 2006, the entirety of which is hereby incorporated by reference.
(135) The operation of the device **100**, described above, according to one embodiment is illustrated in sequence in FIGS. **10**A-**10**L in conjunction with a procedure for closing a patent foramen ovale (PFO) in a patient's heart. As shown in FIG. **10**A, the distal end of a suturing device **100** is advanced through a venous access, such as the inferior vena cava, into the patient's left atrium and positioned in the tunnel **8** of the PFO between the septum primum **7** and the septum secundum **6**. The suturing device **100** may be advanced over a guidewire **10** or alternatively delivered through a catheter introducer sheath **11** using techniques which are known in the art.
(136) The suturing device **100** is initially positioned with the distal tip extending beyond the tunnel of the PFO, such that the spreader assembly **30**, and thus the suture clasp deployment arms are adjacent the tip of the secundum primum **7**. As shown in FIG. **10**B, the suture clasp arms **31**A and **31**B may then deployed from the spreader assembly such that the proximal suture clasp arm **31**A extends around the tip of the secundum primum **7**. The suture clasp arm **31**A holds a suture portion **52**A extending from opening **42** (FIG. **2**B) on the suturing device in suture clasp **33**A such that the suture portion is positioned on the opposite side of the secundum primum **7** relative to the suturing device **100**. The distal suture clasp arm **31**B is also extended from the spreader assembly and abuts the septum secundum **6**, causing the suturing device **100** and proximal suture clasp arm **31**A to be pushed toward the septum primum, thus assisting to properly position the proximal suture clasp arm **31**A adjacent to the septum primum **7**.
(137) Once the suture clasp arm **31**A has been properly positioned around the septum primum **7**, needle **161** may be deployed from the suturing device **100** to penetrate the septum primum **7** and engage the suture clasp arm positioned on the opposite side of the septum primum **7**, as shown in FIG. **10**C. As discussed above, the needle **161** is advanced through a passageway in the suturing device and deflected by needle guide **60**A along an angle that intersects the deployed suture clasp arm **31**A as it exits the suturing device **100**. The needle has a sharp distal tip which penetrates the tissue of the septum primum and engages suture clasp **33**A located on the tip of the suture clasp

arm **31**A. The needle is initially advanced through the suture clasp **33**A which holds a suture portion **52**A. When the needle is advanced through the suture clasp **33**A, a groove on the needle tip engages the suture portion **52**A.

(138) As shown in FIG. **10**D, once the suture loop has been engaged, the needle **161** and engaged suture portion **52**A are then retracted through the tissue of the septum primum **7** and into the needle passageway of the suturing device **100**. Once the suture has been engaged and pulled through the tissue of the septum primum, the suture clasp arms **31**A and B may then be closed and the device may be repositioned such that the spreader assembly **30** and suture clasp arms **31**A and B are adjacent the tip of the septum secundum **6**. Alternatively, the suture clasp arms may remain extended.

(139) As shown in FIG. **10**E, the suturing device is withdrawn proximally through the tunnel of the PFO **8** until the suture clasp arms can be deployed such that the distal suture clasp arm **31**B extends around the tip of the septum secundum **6**. The suture clasp arm **31**B holds a suture portion **52**B extending from opening **42** on the suturing device **100** in suture clasp **33**B such that the suture portion **52***b* is positioned on the opposite side of the septum secundum **6** relative to the PFO **8** and the suturing device **100**. The proximal suture clasp arm **31**A is also extended from the spreader assembly and abuts the septum primum **7**, causing the suturing device **100** and distal suture clasp arm **31**B to be pushed toward the septum secundum **6**, thus assisting to properly position the distal suture clasp arm **31**B around the septum secundum **6**.

(140) Once the suture clasp arm **31**B and suture portion **52**B have been properly positioned around the septum secundum **6**, a needle **165** may be deployed from the distal end of the suturing device **100** to penetrate the septum secundum **6** and engage the suture portion **52**B. As shown in FIG. **10**F, the tip of the needle **165** is advanced from a location distal the suture clasp arms **31**A and **31**B through the tissue of the septum secundum **6** towards deployed suture clasp arm **31**B. As discussed above with respect to FIGS. **8**A-**8**C, the needle **165** comprises a turned portion **166**, such that the turned portion **166** will be advanced toward the deployed suture arm **31**B, as shown, when the proximal portion of the needle **165** is pulled proximally through the suturing device **100**. As the turned portion of the needle **166** is advanced from the suturing device **100**, the turned portion **166** is deflected outward by a needle guide **60**B along an angle that intersects the suture clasp **33**B located on the tip of deployed suture arm **31**B. The needle **165** has a sharp distal tip which penetrates the tissue of the septum secundum **6** and engages suture clasp **33**B located on the tip of the suture clasp arm **31**B. The needle **165** is initially advanced through the suture clasp **33**B which holds a portion of suture **52***b*. When the needle **166** is advanced through the suture clasp **33**B, a groove on the needle tip engages the suture portion **52***b*.

(141) As shown in FIG. **10**G, once the suture portion **52**B has been engaged, the needle **165** and engaged suture portion **52**B are then retracted through the tissue of the septum secundum **6** and into a cavity on the distal tip **70** of the suturing device **100**. Once the suture portion **52**B has been engaged and pulled through the tissue of the septum secundum, the suture clasp arms **31**A and B may then be closed and the suturing device may be withdrawn from the patient's heart. As shown in FIG. **10**H, suture portion **52**A has been positioned through the septum primum **7** while suture portion **52**B has been positioned through the septum secundum. After the suturing device **100** has been withdrawn, the suture portions **52**A and **52**B will extend proximally from the PFO tunnel **8**. These suture end portions can then be pulled tight as shown in FIG. **10**I to draw the septum secundum **6** and septum primum **7** towards one another and close the PFO.

(142) FIG. **10**J shows an alternative configuration for the septum primum and the septum secundum after the suture end portions are pulled tight to close the PFO. In FIG. **10**J, the flap of the septum secundum **6** turns or folds so that the tips of both the septum primum and septum secundum extend in the same direction. FIG. **10**K illustrates an alternative embodiment where the flap of the septum primum turns or folds so that the tip of the septum primum extends in the opposite direction compared to the tip of the septum secundum.

(143) With the suture portions **52**A and **52**B extending away from the PFO, a knot may be applied to the PFO to close the PFO. For example, a device for applying a knot may be used, such as described in U.S. Patent Publication No. 2007-0010829 A1, published Jan. 11, 2007, the entirety of which is hereby incorporated by reference. FIG. **10**L illustrates another embodiment in which a patch **254** may be applied and may be delivered over the suture portions **52**A and **52**B to the PFO. Further details regarding delivery of a patch, as well as other devices, structures and methods that may be incorporated with the above or below embodiments, may be found in U.S. Pat. Nos. 5,860,990, 6,117,144, and 6,562,052, the entirety of each which is hereby incorporated by reference.
(144) FIGS. **11**A-**11**E illustrate an alternative sequence for delivering suture to close a PFO. In FIG. **11**A, the extended suture clasp arm **31**B is first positioned around the septum secundum **6**, and as shown in FIG. **11**B, the needle **165** is advanced proximally through tissue of the septum secundum to engage the suture portion **52**B. The needle is withdrawn from the septum secundum **6**, carrying the suture portion **52**B with it, into the elongate body. In FIG. **11**C, the extended suture clasp arm **31**A is then positioned around the septum primum **7**, and as shown in FIG. **11**D, the needle **161** pierces the tissue of the septum primum to engage suture portion **52**A in suture clasp arm **31**A. As shown in FIG. **11**E, the needle **161** is retracted, carrying the suture portion **52**A into the elongate body. The suturing device **100** can then be withdrawn from the PFO, and the PFO can be closed as described above.
(145) FIGS. **12**A-**12**L illustrate another embodiment of a suturing device **100** used for closing the PFO. The suturing device **100** is similar to the suturing devices described above, except that the distal tip **70** of the suturing device may comprise an elongated distal tip that assists in navigating the device over the guidewire **10** and through the tunnel of the PFO. As illustrated in FIG. **12**A, the suture portions **52**A, **52**B may exit the port or opening **42**, with suture portion **52**B extending to the distal suture clasp arm **31**B, and the suture portion **52**A extending into port **71** before returning proximally to the proximal suture clasp arm **31**A. The suturing device **100** may be delivered over the guidewire **10** to the PFO as described above, and as illustrated in FIG. **12**A, may be positioned with the spreader assembly **30** located proximal to the PFO tunnel **8**.
(146) As shown in FIG. **12**B, the suture clasp arms **31**A and **31**B may be deployed from the spreader assembly **30**. The spreader assembly **30** may be spaced away from the PFO prior to deployment of the suture clasp arms, to allow the arms room to deploy. The suturing device **100** may then be advanced such that the distal suture clasp arm **31**B engages or is positioned adjacent the septum secundum **6**. The suture clasp arm **31**B holds suture portion **52**B extending from opening **42** on the suturing device. As shown in FIG. **12**C, needle **165** may then be deployed from the suturing device, passing through tissue of the septum secundum into engagement with suture portion **52**B carried by the suture clasp arm **31**B. Retraction of the needle **165**, as shown in FIG. **12**D, carries the suture portion **52**B through the tissue of the septum secundum **6** and into the body of the suturing device **100**.
(147) With the first suture portion **52**B extending through tissue of the septum secundum, the suturing device may be partially withdrawn from the tunnel of the PFO and the suture clasp arms **31**A and **31**B may be retracted back into the suturing device, as shown in FIG. **12**E. While in this low profile configuration, the suturing device can then be advanced further through the tunnel of the PFO into the left atrium, and the suture clasp arms **31**A and **31**B can be deployed once they are positioned past the tunnel of the PFO, as shown in FIG. **12**F. FIG. **12**G shows the suturing device **100** partially retracted to cause the proximal suture clasp arm **31**A to engage or be positioned around or adjacent the septum primum **7**. As shown in FIG. **12**H, needle **161** can then be advanced from the suturing device, through tissue of the septum primum **7**, and into engagement with the suture portion **52**A carried by the suture clasp arm **31**A. Retraction of the needle **161** back into the suturing device, as shown in FIG. **12**I, carries the suture portion **52**A through tissue of the septum primum **7**.

(148) FIG. **12**J illustrates the suturing device advanced further into the left atrium to allow the suture clasp arms **31**A and **31**B room to close before withdrawal of the device. With the arms closed or retracted, as shown in FIG. **12**K, the suturing device **100** can be withdrawn from the tunnel of the PFO, as shown in FIG. **12**L. The PFO may be closed using the suture portions **52**A and **52**B according to methods hereinbefore described.
(149) FIG. **13** shows a suturing device **1100** for suturing an opening in a vessel wall or other biological tissue. While the device will be described in reference to suturing an opening in the heart wall, such as a patent foramen ovale (PFO), the device **1100**, like the device **100**, could be used to close other openings in the heart wall, such as a patent ductus arteriosus (PDA) or an atrial septal defect (ASD), other openings in bodily tissue, or the like. The device **1100** could also be used to suture adjacent biological structures or any other time it may be desired to apply a suture to a biological structure, or to perform other procedures as described above with respect to the device **100**.
(150) The suturing device **1100** comprises an elongate tubular member **1020** having a spreader assembly **1090**, shown in greater detail in FIGS. **14**A and **19**, connected to the distal end of the elongate tubular member **1020** for positioning in the opening of the PFO. A handle **1200** is provided at the proximal end of the tubular member **1020**. The elongate tubular member **1020** can be similar to the elongate tubular member **20** in some respects. For example, as shown in more detail in FIG. **15**A, the elongate tubular member **1020** has a plurality of lumens **1021**, **1022**, **1023** and **1024** extending along the axial length of the member **1020** in a generally stacked arrangement. Also, the elongate tubular member **1020** can have similar dimensions to the elongate tubular member **20**. Additionally, the member **1020** can be manufactured by similar techniques and from similar materials to the member **20**.
(151) The elongate tubular member **1020** can differ from the elongate tubular member **20** in some respects. For example, in addition to the lumens **1021**, **1022**, **1023** and **1024** arranged generally across a diameter of the elongate tubular member **1020**, the elongate tubular member **1020** may additionally comprise a one or more lumens, such as lumens **1025**, **1026**, and **1027** that are shown in FIG. **15**A as extending axially within elongate tubular member **1020** on either side of the lumens **1021**, **1022**, **1023** and **1024**.
(152) In some embodiments, as will be discussed below in more detail, the lumens **1021**, **1022**, **1023**, **1024**, **1025**, **1026** and **1027** can be used to provide access through the elongate tubular member **1020** for a guidewire, to provide access for one or more actuating rods connected to suture clasp arms, to house one or more suture catch mechanisms or needles, and to deliver one or more sutures to the distal end of the elongate tubular member **1020**. Some embodiments may also employ these lumens, or include further lumens, for injection of die, housing an additional guidewire, or to facilitate molding of the elongate tubular member.
(153) For example, an elongate tubular member **1020**′, shown in FIGS. **14**B and **15**B, includes lumens for injection of die and an additional guidewire. The elongate tubular member **1020**′ is similar to the elongate tubular member **20** in some respects. Thus, similar features of the elongate tubular member **1020**′ are indicated with numerals similar to those of elongate tubular member **1020** followed by a prime ('). Additionally, elongate tubular member **1020**′ comprises a lumen **1028**′ through which die can be injected into the treatment site, and a lumen **1029**′ for a second guidewire **1010**. In additional embodiments, further lumens may be used for other purposes, such as for injection of die to an additional location or to house yet another guidewire.
(154) The elongate tubular member **1020**′ can comprise one or more openings near the spreader assembly **1090** between the lumen **1028**′ and the treatment site to permit expulsion of die from the lumen **1028**′. Alternatively or additionally, die may pass from the lumen **1028**′ through a spreader assembly **1090**′ to the treatment site.
(155) Similarly, the second guidewire **1010** can pass through the lumen **1029**′ into the spreader assembly **1090**′, which can have a guide, ramp or other feature to direct the second guidewire **1010**

through an opening **1043′** and away from the spreader assembly **1090′**. Alternatively or additionally, the elongate tubular member **1020′** can comprise an opening proximal to and near the spreader assembly **1090′** between the lumen **1029′** and the treatment site for passage of the second guidewire **1010**.
(156) Referring again to FIG. **14**A, the spreader assembly **1090** is bonded, or otherwise joined, to the distal end of the elongated tubular member **1020**, for example, with epoxy or by any other suitable technique known to those skilled in the art. The spreader assembly **1090** can comprise a spreader **1030** (FIG. **18**), one or more suture clasp arms **1031**, a casing **1040**, and a distal tip **1070**.
(157) Referring to FIGS. **16-18**, the spreader **1030** comprises one or more lumens. The spreader **1030** has lumens **1133** and **1136** that extend generally parallel to a longitudinal axis of the spreader assembly **1090**, as shown in FIG. **16**. When the spreader **1030** and the elongate tubular member **1020** are properly connected, the lumens **1022**, **1023**, and **1024** of the member **1020** are preferably substantially aligned with the lumen **1133** of the spreader **1030**, and the lumen **1026** of the member **1020** is preferably substantially aligned with the lumen **1136** of the spreader **1030** to provide continuous passageways between the elongate tubular member **1020** and the spreader **1030**.
(158) In some embodiments, a casing **1040** is placed over the connection between the elongate tubular member **1020** and the spreader **1030** to facilitate proper alignment of the internal lumens of the elongate tubular member **1020** and spreader **1030**, as shown in FIG. **18**A. The casing is preferably made of metal, but can also be made of other materials such as plastics.
(159) As illustrated in FIG. **14**A, the casing **1040** can comprise openings, or recesses, **1041**A and **1041**B located on the opposite sidewalls of the casing **1040** for allowing the release and deployment of a pair of suture clasp arms **1031**A and **1031**B housed within the spreader **1030**. Accordingly, the openings **1041**A and **1041**B are sized and shaped to permit the suture clasp arms to fully extend from the spreader **1030**.
(160) In some embodiments, the casing **1040** has a length such that when the proximal end of the casing **1040** is positioned over the connection between the spreader **1030** and the elongate tubular member **1020**, the distal end of the casing **1040** extends beyond the distal end of the spreader **1030** and, in some embodiments, engages a distal tip **1070** and/or a housing, such as housing **1080** shown in FIGS. **18**A and **19**, that is positioned between the spreader **1030** and the tip **1070**.
(161) As shown in FIG. **14**A, the pair of suture clasp arms **1031**A and **1031**B are housed in the recesses **1041**A and **1041**B in the spreader assembly **1090**. During storage and delivery of the suturing device, the suture clasp arms **1031**A and **1031**B are situated substantially parallel to the longitudinal axis of the suturing device such that the outer walls of the suture clasp arms do not extend beyond the outer diameter of the spreader assembly **1090**.
(162) The suture clasp arms **1031**A and **1031**B are connected to actuating rods **1035**A and **1035**B which extend through the passageways formed by lumens **1133** and **1136** in the spreader assembly and lumens **1024** and **1026** in the elongate tubular member **1020**, as shown in FIG. **15**A. After insertion and positioning of the suturing device at the PFO or other opening, the arms **1031**A and **1031**B can be deployed to the position shown in FIG. **19** by movement of the actuating rods **1035**A and **1035**B relative to the spreader **1030**.
(163) In the embodiment illustrated in FIGS. **14**A and **19**, the distal suture clasp arm **1031**A and the proximal suture clasp arm **1031**B are manufactured as separate components. The distal suture clasp arm **1031**A and the proximal suture clasp arm **1031**B are shown in greater detail in FIGS. **20** and **21**, respectively. Each of the arms **1031** may be pivotally connected to the spreader **1030** at a first connection point, such as pivot apertures **1091**, and connected to the actuating rods **1035** at a second connection point, such as apertures **1092**. The arms may be connected to the spreader **1030** and the actuating rods **1035** by pins or by other techniques, such as by integral construction employing one or more compliant hinges. Accordingly, movement of the actuating rods **1035** relative to the spreader **1030** results in movement of the arms **1031**.
(164) When deployed, the suture clasp arms **1031**A and **1031**B extend from the suturing device

1100 in opposite directions from the longitudinal axis of the device. Preferably, the arms **1031**A and **1031**B form an acute angle with the longitudinal axis of the spreader. In some embodiments, a first arm **1031**A extends outward toward the proximal end of the spreader assembly **1090** at an angle of between about 35-55.degree., alternatively about 40-50.degree., alternatively about 45.degree. with respect to the longitudinal axis of the spreader assembly **1090**. The second arm **1031**B can extend outward toward the distal end of the spreader assembly **1090** at approximately the same angle as the first arm **1031**A with respect to the longitudinal axis of the spreader assembly **1090**. In other embodiments, the second arm **1031**B can extend outward from the spreader assembly **1090** at angle larger or smaller than the angle of the first arm **1031**A with respect to the longitudinal axis of the spreader assembly **1090**.
(165) As shown in FIGS. **29**B and **29**F, the suture clasp arms **1031**A and **1031**B can be independently actuated to be individually deployed depending upon the location of the tissue portion to be sutured. For example, in closing a PFO, the suturing device will first be positioned with respect to a first tissue portion to be sutured and then moved to a second position for suturing a second tissue portion. More specifically, as shown in FIG. **29**B, when closing a PFO, the suturing device **1100** can be first positioned such that suture clasp arm **1031**A extends around the septum primum **7** such that suture portion **52**A may be engaged and pulled though the septum primum **7** to draw it toward the septum secundum **6**. The non-engaged suture clasp arm **1031**B can remain retracted in the spreader **1030**, as shown in FIG. **29**B.
(166) The suture clasp arms **1031**A, **1031**B, like the suture clasp arms **31**, have suture clasps **1033**A, **1033**B (see FIG. **19**) for receiving and holding a suture **50**. The suture clasps **1033**A, **1033**B of the suture clasp arms **1031**A, **1031**B may be configured in a manner similar to or the same as that described with respect to the suture clasps **33**. For example, each clasp **1033** can be a circular opening with a diameter sized to securely receive and hold a loop of suture **50**.
(167) In use, the suture ends are positioned in the suture clasps **1033**A and **1033**B of the suture clasp arms **1031**A and **1031**B and held securely there until they are engaged and removed by suture catch mechanisms **1161** and **1165**. As discussed above, the suture clasp arms **1031**A and **1031**B can comprise slots **1034**A and **1034**B (FIGS. **20** and **21**) to guide the suture into the suture clasps **1033** at the proper angle and assist in maintaining the suture loops in the suture clasp until they are engaged by the suture catch mechanisms. When the suture catch mechanism engages the loops of the suture ends, the suture loops will slide out of the suture clasp **1033** and be released by the suture clasp arm **1031**.
(168) In some embodiments, the suture clasp arms **1031** can comprise legs **1093** that extend away from the clasps **1033**, as illustrated in FIGS. **20** and **21**. The legs **1093** can engage tissue portions at a location remote from a location where a suture catch mechanism is to penetrate a biological structure. The legs **1093** can gather the tissue toward the location where the suture catch mechanism is to penetrate the biological structure to facilitate secure suturing of the biological structure.
(169) The suture **50** can be housed in the lumen **1025** (see FIG. **15**A) of the elongate tubular member **1020**. The suture lumen **1025** can have a port or opening **1042** near the distal end of the elongate tubular member **1020**, as shown in FIG. **19**. The suture **50** can be advanced through the opening **1042** such that the suture ends, or end portions **52**A and **52**B, extend outside of the suturing device **1100**. One of the suture end portions **52**B may extend from the opening **1042** to the suture clasp arm **1031**B. The other suture end portion **52**A may extend along the length of the casing and into a side opening **1081** in the housing **1080** (see FIGS. **19**, **24**, and **26**). The suture portion **52**A may loop back out of the opening **1081** to the suture clasp arm **1031**A.
(170) The housing **1080** can have one or more openings that align with complementary openings in the spreader **1030**. For example, the housing **1080** can comprise a groove, slot or lumen **1089** extends generally parallel to a longitudinal axis of the housing **1080**, as shown in FIGS. **24** and **25**. When the spreader **1030** and the housing **1080** are properly connected, the slot **1089** of the housing

1080 is preferably substantially aligned with a groove, slot, or lumen **1139** (see FIGS. **16-17**) of the spreader **1030** to provide continuous passageways between the housing **1080** and the spreader **1030**. In some embodiments, the housing **1080** can comprise an aperture **1085** that intersects the slot **1089** and extends distally toward a central longitudinal axis of the housing **1080**, as shown in FIG. **25**.
(171) Additionally or alternatively, the housing **1080** can comprise apertures **1083** and **1088**, shown in FIG. **24**, that extend generally parallel to a longitudinal axis of the spreader assembly **1090**. When the spreader **1030** and the housing **1080** are properly connected, the apertures **1083** and **1088** of the housing **1080** are preferably substantially aligned with lumens **1133** and **1138** (see FIG. **17**) of the spreader **1030** to provide continuous passageways between the housing **1080** and the spreader **1030**.
(172) The aperture **1088** can extend generally along the longitudinal axis of the housing **1080** and may extend entirely through the housing or may, as shown in FIG. **24**, extend only partially through the housing **1080**. The aperture **1080** can be intersected by the side opening **1081** and an opening **1082**. In some embodiments, the aperture **1083** can be intersected by an aperture **1084**.
(173) In some embodiments, the housing can comprise a suture retention mechanism. For example, referring to FIG. **26**, the aperture **1088** can accommodate a follower pin **1094** and a spring **1095** that urges the follower pin out of the opening **1088**. A cross pin **1096** can extend through the opening **1082** and engage the follower pin **1094** to limit the range of travel of the follower pin **1094** and inhibit rotation of the follower pin **1094** within the aperture **1088**.
(174) A proximal end of the follower pin **1094** engages a cam leg **1097** of the arm **1031**A, shown in FIG. **20** such that when the arm **1031**A is in the retracted position (see FIG. **14**A) the cam leg **1097** pushes the follower pin **1094** into the aperture **1088** compressing the spring **1095**. As the arm **1031**A is extended from the spreader **1030** the cam leg **1097** moves away from the housing **1080** allowing the follower pin **1094** to move proximally.
(175) The follower pin **1094** can have a hook **1098**, as illustrated in FIG. **26**. A portion of the suture **50** that is looped through the opening **1081** extends around the hook **1098**. When the arm **1031**A is moved into the closed position, the follower pin **1094** moves distally such that the hook retains the suture **50** within the opening **1081**. As the follower pin **1094** moves proximally when the arm **1031**A is extended from the spreader **1030**, the hook **1098** releases the suture **50** to permit the suture **50** to be pulled from the opening **1081**.
(176) The suture retention mechanism can allow the user of the device **1100** to control the release of the suture **50** from the opening **1081** in the housing **1080** to reduce the likelihood of the suture end portion **52**A being pulled prematurely from the suture clasp **1033**A, to provide slack in the suture **50** when the suture catch mechanism retrieves the suture end portion **52**A, and to release the suture **50** to permit the suture catch mechanism pull the suture **50** through a biological structure.
(177) In some embodiments, the suture catch mechanism **1161** can be similar to or the same as the suture catch mechanism **161**, shown in FIG. **7**A. The suture catch mechanism **1161** can be configured for engaging the suture clasp arm **1031**A and can comprise an elongate, straight needle.
(178) In some embodiments, the suture catch mechanism **1165** can be similar to or the same as the suture catch mechanism **165**, shown in FIG. **7**B. The suture catch mechanism **1165** is configured to engage a suture clasp arm **1031**B and can comprise a bent needle.
(179) In use, the needles **1161** and **1165**, like the needles **161** and **165** shown in FIG. **8**A, can be housed within passageways formed by lumens within the elongate tubular member **1020**, spreader **1030**, and the housing **1080**. Referring to FIG. **18**A, the suture catch mechanism **161** is slidably housed in the lumen **1021** of the elongate tubular member **1020**, while suture catch mechanism **165** is slidably housed in the passageway formed by the lumen **1023** of the elongate tubular member, the lumen **1133** of the spreader **1030**, and lumen **1083** of the housing **1080**.
(180) As shown in FIGS. **16-18**, the spreader **1030** includes a plurality of needle guides **1060**A and **1060**B for guiding a plurality of suture catch mechanisms, such as a needle or other penetrating

mechanism, towards the deployed suture clasp(s). In one embodiment, the spreader **1030** may comprise two needle guides **1060**A and **1060**B located on the distal and proximal ends of the spreader **1030** for guiding two needles toward deployed suture arms **1031**A and **1031**B illustrated in FIG. **19**. Each of the needle guides **1060**A and **1060**B comprises an angled groove or channel in the sidewall of the spreader **1030** such that it will deflect a suture catch mechanism, or needle, exiting the suture assembly **1090** along a path that intercepts the suture clasps **1033**A and **1033**B of the suture clasp arms when the suture clasp arms **1031**A and **1031**B are in a deployed position. In some embodiments, the suture arms **1031**A and **1031**B may be similar to or the same as the suture arms **31**A and **31**B. For example, the suture arms **1031**A and **1031**B can be sized such that a groove having an angle of between about 10-35.degree., alternatively about 15-25.degree., alternatively about 19.degree. with respect to the longitudinal axis of the spreader assembly will spread the needles to the proper angle to engage the suture loops in the deployed suture clasps **1033**.
(181) In use, the proximal needle guide **1060**A (FIGS. **16** and **18**A) is aligned with the lumen **1021** of the elongate tubular member such that when a suture catch mechanism is advanced through the lumen **1021**, the distal end of the needle will be deflected by the needle guide **1060**A towards the deployed suture clasp arm **1031**A. Once the needle **1161** penetrates the suture clasp **1033**A on the suture clasp arm **1031**A and engages the suture portion **52**A held by the suture clasp **1033**A, the needle **1161** may be retracted back into the lumen **1021** with the suture portion **52**A held on the distal end of needle **1161**, in a manner similar to that shown in FIG. **8**C.
(182) Likewise, distal needle guide **1060**B (FIGS. **17** and **18**A) is aligned with aperture **1083** of the housing **1080** such that when the distal tip of a needle housed in aperture **1083** is extended towards the suture clasp arm **1031**B, the needle will be deflected toward the suture clasp **1033**B. Before insertion, the needle **1165**, similar to the needle **165**, has been advanced through the lumen **1023** in the elongate tubular member **1020** beyond the spreader **1030** and positioned a slot or cavity **1083** within the housing **1080** of the suturing device that has been aligned with the lumen **1133** of the spreader **1030**. The cavity **1083** is sized to receive the distal end of the needle **1165**, including a turned portion similar to the turned portion **166** (see FIG. **7**B). The turned portion is positioned in cavity **1083** such that the distal tip is aligned with the needle guide **1060**B located on the distal end of the spreader **1030**.
(183) Thus, in use, when the proximal end of the needle **1165** is pulled back through the lumen **1023** of the elongate tubular member **1020**, the turned portion of the needle will be advanced along the groove of the needle guide **1060**B and deflected outward along the angle path of the groove to penetrate the deployed suture clasp **1033**B on the suture clasp arm **1031**B and engage suture portion **52**B held therein. Once the needle **165** has engaged the suture portion **52**B, the proximal end of the needle **165** may then be pushed forward through the lumen **1023** of the elongate tubular member which will cause the bent portion of the needle to be retracted along needle guide **1060**B into cavity **1083** along with the suture portion **52**B held on the needle **1165**.
(184) Under some circumstances, a suture catch mechanism, such as the needle **1161** or the needle **1165**, when advanced may be oriented in direction toward a location slightly closer to or slightly farther from a longitudinal axis of the device **1100** than the center of the suture clasp **1033**. A suture catch mechanism that is oriented in a direction that is toward a location slightly closer to or slightly farther from the longitudinal axis of the device **1100** than the center of the suture clasp **1033** may, in some instances, not properly engage the suture. As a result, the suture catch mechanism may not successfully retract the suture end portion.
(185) In some embodiments, the one or more of the suture clasp arms **1031** can have a deflector, such as deflecting plates **1099**, **2099**, shown in FIGS. **20-23**. The deflecting plates **1099**, **2099** are shown partially obscuring an opening on a back side of the suture clasp **1033** relative to the direction from which a suture catch mechanism, such as a needle **2161**, penetrates the suture clasp **2033**, as shown in FIG. **22**. As the needle **2161** passes through the suture clasp **2033**, the needle **2161** engages the deflector plate **2099**, as shown in FIG. **22**, and is diverted from its previous

course toward a longitudinal axis of the device **1100**. In an alternative embodiment, the suture catch mechanism can be diverted away from a longitudinal axis of the device. As the needle **2161** is diverted toward the longitudinal axis of the device, a suture engaging portion, such as a hook, recess, or groove **2161**, moves toward a portion of a suture end **2052**, shown schematically in FIG. **22**.
(186) The dimensions of the needle including the size and location of the suture engaging portion and the size and shape the needle tip, the size of the suture clasp, and the distance by which the deflectors obscure the back opening of the suture clasp should be relatively proportioned such that the deflectors **1099**, **2099** cause the suture engaging portion of the needle to engage the portion of the suture end as the needle returns to its previous orientation. Thus, in the embodiment illustrated in FIG. **22**, the hook **2162** would engage the suture portion **2052** as it retracted from the suture clasp **2033**.
(187) The deflector plates may be generally rectangular, as shown in FIG. **23**, or may have other configurations, such as generally H-shaped as shown in FIG. **21**. In some embodiments, the deflector plates can be made from metal, while in other embodiments the deflector plates can be made of plastics or other materials of sufficient rigid or resiliency to deflect a suture catch mechanism. The deflector plates may be joined to the suture clasp arms by welding, epoxy, adhesives or by other methods. In an alternative embodiment, the deflectors can be integrally formed with the suture clasp arm.
(188) In some embodiments, the deflectors can compensate for misalignment of a suture catch mechanism with the center of a suture clasp relative to a longitudinal axis of the device to provide consistent capture of a suture end portion by the suture catch mechanism.
(189) In some embodiments, bending of the elongate tubular member can affect the relative positions of the ends of the suture catch mechanisms to the spreader assembly. For example, if an elongate tubular member is bent then a distal end of a suture catch mechanism extending along the inside of the bend in the elongate member relative to the central axis of the elongate tubular member would be advanced relative to the spreader assembly. In such a circumstance, the suture catch mechanism may be advanced through a suture clasp farther than is desired, which may result in enlargement of a loop at an end of a suture that in turn inhibits the ability of the suture catch mechanism to retract the and of the suture.
(190) On the other hand, if an elongate tubular member is bent then a distal end of a suture catch mechanism extending along the outside of the bend in the elongate member relative to the central axis of the elongate tubular member would be retracted relative to the spreader assembly. In such a circumstance, the suture catch mechanism may not be advanced through a suture clasp far enough to engage a suture end portion held by the suture clasp.
(191) In some embodiments, the effects of bending of the elongate tubular member can be reduced or eliminated by using a suture catch mechanism that is sufficiently long to engage the corresponding suture clasp even if the suture catch mechanism extends along the outside of a bend in the elongate member relative to the central axis of the elongate member. Advancement of the suture catch mechanism may be limited by providing a stop mechanism in proximity to the spreader assembly where the effects of bending are small or absent.
(192) For example, as illustrated in FIG. **18**B, a tube **1061** can be attached to the needle **1161** and another tube **1062** can be attached to the elongate tubular member **1020** within a counterbore in the lumen **1021**. The inner diameter of tube **1062** is smaller than the outer diameter of tube **1061** such that tube **1061** cannot be advanced through tube **1062**, thereby limiting the advancement of the needle **1161** relative to the spreader **1030**.
(193) Alternatively, the tube **1061** can be attached to the needle **1161** and extend over a portion of the length of the needle **1161** within the lumen **1021**. In this configuration, the tube **1061** limits the movement of the suture catch mechanism by preventing movement of the suture catch mechanism along the guide **1060**A by interference of the tube **1061** with the proximal end of the spreader **1030**.

(194) Similarly, movement of the bent needle **1165** toward the suture clasp arm **1031**B can be limited by attachment of a tube **1065** to the needle **1165** at a location distal to the spreader **1030**. The tube **1065** has an outer diameter that is larger than a dimension of the lumen **1133** such that the tube **1065** cannot be moved into the spreader **1030** as the needle **1165** is moved proximally, thereby limiting the movement of the needle **1161** relative to the spreader **1030** in a proximal direction.
(195) In an alternative embodiment, the suture catch assembly may comprise four needles, for example two needles housed on each side of the elongate tubular member for engaging multiple suture portions in each of the suture clasp arms. Here, each of the needles may be independently actuated or alternatively both needles for engaging a single suture clasp arm may be jointly actuated such that they are deployed at the same time.
(196) The distal tip **1070** is shown in FIGS. **14**A, **19**, and **29**A as having a tapered configuration, but may alternatively be rounded or have other configurations. The distal tip **1070** is preferably atraumatic and can be bonded, adhered, or otherwise joined to the distal end of the spreader assembly **1090**, for example, by insert molding or any other suitable technique known to those skilled in the art. The distal tip **1070** can have at least one lumen **1072** that forms a continuous passage with the aperture **1085** and the slot **1089** of the housing **1080** (FIG. **25**) and the slot **1039** (FIG. **16**) of the spreader **1030** and lumen **1027** of the elongate tubular member **1020** (FIG. **15**A), which may accommodate, for example, a guidewire.
(197) In addition, the distal tip **1070** may have one or more additional lumens **1173** that can be aligned with lumens in the spreader assembly **1030** and elongate tubular member **1020** to provide additional continuous passageways, for example for housing a suture catch mechanism. Here, as discussed above, the outer diameter of the distal tip is substantially the same size as the inner diameter of the casing **1040** such that when the distal end of the casing **1040** is positioned over the connection between the distal tip **1070**, the housing **1080**, and the spreader **1030**, the casing maintains the proper alignment of the internal lumens of the distal tip **1070**, the housing **1080**, the spreader **1030**, and the elongate member **1020**.
(198) Referring now to FIGS. **27**, **28**A, and **28**B, the proximal end of the needles **1161** and **1165** extend through the lumens **1021** and **1023** of the elongate tubular member and terminate at a connection to pulls **1210** and **1216** on the suturing device handle **1200**. The handle **1200** includes a housing **1201** which is attached to the proximal end of the elongate tubular member **1020**. The housing **1201** has an aperture **1202** providing a passageway between the handle **1200** and the multiple lumens of the elongate tubular member **1020**. One or more actuators **1210**, **1216**, **1240** and **1260** may extend from the housing **1201**. The pulls or knobs **1210**, **1216**, **1240** and **1260** can be connected to actuating rods or mechanisms for deploying and/or retracting the suture clasp arms and suture catch mechanisms moveably housed in the suturing device.
(199) The elongate member **1020** is fixed to the housing **1201** in any suitable manner such as by welding, adhesion, or other bonding process to a collar **1203**. The cylinder **1203** is engaged by the housing **1201**. The cylinder **1203** can be rectangular or any other shape. The cylinder **1203** is connected to the housing **1201** in a manner that prevents limits rotation of the cylinder **1203** with the member **1020** relative to the housing **1201**. The elongate member **1020** has a plurality of cutouts (not shown) that expose the actuator rods **1035** and the suture catch mechanisms **1161** and **1165** for connection to the actuators **1210**, **1216**, **1240**, and **1260**.
(200) With reference to FIGS. **27**, **28**A, and **28**B, in some embodiments, the handle **1200** can have two pulls **1210** and **1216** for deploying and retracting the suture catch mechanisms such as needles **1161** and **1165** of the suturing device. The pulls **1210** and **1216** can extend from the housing through openings **1213** and **1217**, respectively, as shown in FIG. **27**.
(201) In one embodiment, the proximal end of needle **1165** extends from the proximal end of the lumen **1023** in the elongate tubular member through opening **1202** into the handle **1200** and is exposed by a cutout in the elongate tubular member **1020** for connection to a shaft **1276**, shown in FIGS. **28**A and **28**B. The shaft **1276** can be generally U-shaped to permit the shaft to pass over the

elongate member **1020** in a direction transfers to a longitudinal axis of the member **1020**. The needle **1165** may be connected to the shaft **1276** directly by a suitable bonding process, or indirectly, such by connection to a small tube that is adhered to the needle **1165**, for example, by epoxy, which is in turn connected to a tab **1286**. The tab **1286** can be connected to the shaft **1276** at a notch **1296** in the shaft **1276**. The pull **1216** can be connected to the shaft by a bonding process or by mechanical means, such as a set screw.

(202) Similarly the proximal end of needle **1161** extends from the proximal end of the lumen **1021** in the elongate tubular member **1020** through opening **1202** into the handle **1200** and is exposed by a cutout in the elongate tubular member **1020** for connection to a shaft **1270**, shown in FIGS. **28**A and **28**B. The shaft **1270** can be generally U-shaped to permit the shaft to pass over the elongate member **1020** in a direction transfers to a longitudinal axis of the member **1020**. The needle **1161** may be connected to the shaft **1270** directly by a suitable bonding process, or indirectly, such as by connection to a small tube (not shown) that is adhered to the needle **1161**, for example, by epoxy, which is in turn connected to a tab **1280**. The tab **1280** can be connected to the shaft **1270** at a notch **1290** in the shaft **1270**. The pull **1210** can be connected to the shaft by a bonding process or by mechanical means, such as a set screw.

(203) In some embodiments, a spring **1218** can be positioned between the pulls **1210** and **1216**. The spring **1218** can bias the needle **1161** proximally and the needle **1165** distally by biasing the pull **1210**, which is connected to the needle **1161**, from the pull **1216**, which is connected to the needle **1165**. A user can advance the needle **1161** by moving the pull **1210** to compress the spring **1218**. The spring **1218** can assist in retraction of the needle **1161** as the spring **1218** decompresses. Similarly, a user can draw the needle **1165** proximally by moving the pull **1216** to compress the spring **1218**. As the spring **1218** decompresses the needle **1165** is moved distally within the elongate member **1020**.

(204) The suture claps arms **1031**A and **1031**B can be deployed by rotation of actuators, or knobs, **1240** and **1260**. For example, the knobs **1240**, **1260** can be connected to threaded members **1242**, **1262** in a manner that prevents the knobs **1240**, **1260** from rotating with respect to the threaded members **1242**, **1262**. A tube **1222**, which can be fixedly attached to the housing **1201**, can extend through the threaded members **1242**, **1262** such that the threaded members **1242**, **1262** can both slide over and rotate with respect to the tube **1222**.

(205) The tube **1222** can have two slots **1243** and **1263** corresponding to the threaded members **1242**, **1262**. The two slots **1243** and **1263** are configured to allow two pins **1244** and **1264** to extend through the slots **1243**, **1263** and operatively engage the threaded members **1242**, **1262** such that rotation of the threaded members **1242**, **1262** with respect to the tube **1222** causes the pins **1244**, **1264** to move along the length of the slots **1243**, **1263**.

(206) The pins **1244**, **1264** are connected to shafts **1245**, **1265** positioned within the tube **1222**. The shafts **1245**, **1265** can be connected to the actuator rods **1035**A and **1035**B by tabs, **1246**, **1266**. The tabs **1246**, **1266** can be welded or otherwise attached to the actuator rods **1035**A and **1035**B and attached by any suitable bonding process to the shafts **1245**, **1265** at notches **1247**, **1267**.

(207) Accordingly, as a user rotates knobs **1240**, **1260**, the threaded members **1242**, **1264** move the pins **1244**, **1264** along the slots **1243**, **1263**. Movement of the pins **1244**, **1264** in a direction along a longitudinal axis of the handle **1200** causes the shafts **1245**, **1265** to move the actuator rods **1035**A and **1035**B either proximally or distally depending on the direction of rotation of the knobs **1240**, **1260**. As discussed above, proximal and distal movement of the actuator rods **1035**A and **1035**B causes the suture clasp arms **1031**A and **1031**B to move between a retracted position, shown in FIG. **14**A, and an extended position, shown in FIG. **19**. The knobs **1240**, **1260** can be rotated individually to independently deploy and retract the suture clasp arms **1035**A and **1035**B.

(208) In some embodiments, the handle **1200** can also comprise springs **1248**, **1268** that bias the threaded members **1242**, **1262** along the tube **1222**. The springs **1248**, **1268** can bias the threaded members **1242**, **1262** either proximally or distally. In one embodiment, the springs **1248**, **1268** bias

the threaded members **1242**, **1262** in directions that bias the suture clasp arms **1031**A, **1031**B toward in their extended positions such that rotation of the threaded member **1242**, **1246** past the fully open position of the suture clasp arms **1031**A, **1031**B results in compression of the springs **1248**, **1268**. Alternatively, the springs **1248**, **1268** can bias the threaded members **1242**, **1262** in directions that bias the suture clasp arms **1031**A, **1031**B toward in their retracted positions.
(209) The operation of the device **1100**, described above, is illustrated according to one embodiment in FIGS. **29**A-**29**I in conjunction with a procedure for closing a patent foramen ovale (PFO) in a patient's heart. As shown in FIG. **29**A, the distal end of a suturing device **1100** is advanced through a venous access, such as the inferior vena cava, into the patient's left atrium and positioned in the tunnel **8** of the PFO between the septum primum **7** and the septum secundum **6**. The suturing device **1100** may be advanced over a guidewire **10** or alternatively delivered through a catheter introducer sheath **1011** using techniques which are known in the art.
(210) The suturing device **1100** is initially positioned with the distal tip **1070** extending beyond the tunnel of the PFO, such that the spreader assembly **1090** is near the tip of the secundum primum **7** and at least the suture clasp deployment arm **1031**A is permitted to extend from the spreader assembly **1090**. The suture clasp arm **1031**A may then deployed from the spreader assembly **1090** and then the device **1100** is retracted until the suture clasp arm **1031**A extends around the tip of the secundum primum **7**, as shown in FIG. **29**B, and gathers the tissue of the septum primum **7** between the arm **1031**A and the spreader **1030**. Referring to FIG. **26**, as the arm **1031**A is extended, the follower pin **1094** moves proximally to permit the suture **50** to be pulled from the opening **1081** in the housing **1080**. With continued reference to FIG. **29**B, the suture clasp arm **1031**A holds a suture portion **52**A, which may or may not still extend from opening **1081** of the housing **1080**, in the suture clasp **1033**A such that the suture portion **52**A is positioned on the opposite side of the secundum primum **7** relative to the suturing device **1100**.
(211) Once the suture clasp arm **1031**A has been properly positioned around the septum primum **7**, needle **1161** may be deployed from the suturing device **1100** to penetrate the septum primum **7** and engage the suture clasp arm positioned on the opposite side of the septum primum **7**. As discussed above, the needle **1161** is advanced through a passageway in the suturing device and deflected by needle guide **1060**A (FIG. **18**A) along an angle that intersects the deployed suture clasp arm **1031**A as it exits the suturing device **1100**. The needle has a sharp distal tip which penetrates the tissue of the septum primum **7** and engages suture clasp **1033**A located on the tip of the suture clasp arm **1031**A, as shown in FIG. **29**C. The needle is initially advanced through the suture clasp **1033**A which holds a suture portion **52**A. When the needle is advanced through the suture clasp **1033**A, a groove or hook on the needle tip engages the suture portion **52**A.
(212) As shown in FIG. **29**D, once the suture loop has been engaged, the needle **1161** and engaged suture portion **52**A are then retracted through the tissue of the septum primum **7** and into the lumen **1021** of the elongate member **1020** of the suturing device **1100**. Once the suture has been engaged and pulled through the tissue of the septum primum **7**, the device **1100** may be advanced slightly so that the suture clasp arm **1031**A can be closed without pinching the septum primum **7**. The device **1100** may then be repositioned such that the spreader assembly **1090** is adjacent the tip of the septum secundum **6**.
(213) As shown in FIG. **29**E, the suturing device **1100** is withdrawn proximally through the tunnel of the PFO **8** until the suture clasp arm **1031**B can be deployed. The suture clasp arm **1031**B can then be extended then the device **1100** can be advanced such that the suture clasp arm **1031**B extends around the tip of the septum secundum **6**, as shown in FIG. **29**F, and gathers the tissue of the septum secundum **6** between the arm **1031**B and the spreader **1030**. The suture clasp arm **1031**B holds a suture portion **52**B extending from opening **1042** in suture clasp **1033**B such that the suture portion **52**B is positioned on the opposite side of the septum secundum **6** relative to the PFO **8** and the suturing device **1100**.
(214) Once the suture clasp arm **1031**B and suture portion **52**B have been properly positioned

around the septum secundum **6**, the needle **1165** may be deployed from the distal end of the suturing device **1100** to penetrate the septum secundum **6** and engage the suture portion **52**B. As shown in FIG. **29**G, the tip of the needle **1165** is pulled proximally from a location distal the suture clasp arm **1031**B through the tissue of the septum secundum **6** towards deployed suture clasp arm **1031**B.

(215) The needle **1165** is deflected outward by a needle guide **1060**B (FIG. **18**A) as the needle **1165** extends from the suturing device **1100** along an angle that intersects the suture clasp **1033**B located on the tip of deployed suture arm **1031**B. The needle **1165** has a sharp distal tip which penetrates the tissue of the septum secundum **6** and engages suture clasp **1033**B located on the tip of the suture clasp arm **1031**B. The needle **1165** is initially advanced through the suture clasp **1033**B which holds a portion of suture **52**B. When the needle **166** is advanced through the suture clasp **1033**B, a groove or hook on the needle tip engages the suture portion **52**B.

(216) As shown in FIG. **29**H, once the suture portion **52**B has been engaged, the needle **1165** and engaged suture portion **52**B are then retracted distally through the tissue of the septum secundum **6** and into the cavity **1083** of the suturing device **1100**. The suture clasp arm **1031**B may then be closed and the suturing device **1100** may be withdrawn from the patient's heart.

(217) As shown in FIG. **29**I, the suture portion **52**A has been positioned through the septum primum **7** while suture portion **52**B has been positioned through the septum secundum. After the suturing device **1100** has been withdrawn, the suture portions **52**A and **52**B will extend proximally from the PFO and can then be pulled to draw the septum secundum **6** and septum primum **7** towards one another to close the PFO, as shown in FIG. **10**I, **10**J, or **10**K. As the suture **50** is pulled tight, the suture **50** preferably causes the septum secundum **6** and septum primum **7** to turns or folds so that the tip of the septum primum **7** extends in the opposite direction compared to the tip of the septum secundum **6**, as illustrated in FIG. **10***k*.

(218) With the suture portions **52**A and **52**B extending away from the PFO, a knot may be applied to the PFO to close the PFO. For example, a device for applying a knot may be used, such as described in U.S. Patent Publication No. 2007-0010829 A1, published Jan. 11, 2007, the entirety of which is hereby incorporated by reference. Alternatively, the method of securing the suture portions illustrated in FIG. **10**L by applying a patch **254** may be used.

(219) As noted above, further details regarding delivery of a patch, as well as other devices, structures and methods that may be incorporated with the above or below embodiments, may be found in U.S. Pat. Nos. 5,860,990, 6,117,144, and 6,562,052, the entirety of each which is hereby incorporated by reference.

(220) The operation of the device **1100′**, described above and shown in FIGS. **14**B and **15**B, is illustrated according to one embodiment in FIGS. **29**J-**29**M in conjunction with a procedure for closing a patent foramen ovale (PFO) in a patient's heart. The operation of the device **1100′** can be similar to the operation of the device **1100** in some respects. For example, the distal end of a suturing device **1100′** is advanced through a venous access, such as the inferior vena cava, into the patient's left atrium and positioned in the tunnel **8** of the PFO between the septum primum **7** and the septum secundum **6** then operated in the manner described above with reference to FIGS. **29**A-D to pass a suture portion **52**A through the septum primum **7**.

(221) The operation of the device **1100′** can differ from the operation of the device **1100** in other respects. In particular, the device **1100′** can be used in conjunction with the second guidewire **1010** to facilitate placement of a suture portion through the septum secundum **6** in a location at or near a center of the septum secundum **6**. Such placement of the suture through the septum secundum **6** may improve the likelihood of successful closure of the PFO. If the suture is placed through the septum secundum **6** at a location too far removed from the center of the septum secundum **6**, to one side or the other, blood may continue to flow through the PFO after placement of the suture and application of a knot to the suture. Thus, use of the device **1100′** with the second guidewire **1010** can improve the likelihood of securely closing the PFO.

(222) Once the suturing device **1100**′ has been operated in the manner described above with reference to FIGS. **29**A-D, the suturing device **1100**′ may be withdrawn proximally through the tunnel of the PFO **8** until the suture clasp arm **1031**B′ can be deployed, as shown in FIG. **29**J. The second guidewire **1010** can be advanced from the lumen **1029**′ (FIG. **15**B) through the opening **1043**′ (FIG. **14**B) into the superior vena cava (FIG. **1**). The second guidewire **1010** may be preloaded in the device **1100**′ before introduction of the device **1100**′ into the body such that a distal end of the second guidewire is near the opening **1043**′.
(223) The suture clasp arm **1031**B′ can then be extended and the device **1100**′ can be advanced such that the suture clasp arm **1031**B′ extends around the tip of the septum secundum **6**, as shown in FIG. **29**K, and gathers the tissue of the septum secundum **6** between the arm **1031**B′ and the spreader **1030**′. Alternatively, the suture clasp arm **1031**B′ can be extended from the spreader assembly **1090**′ before the second guidewire **1010** is advanced into the superior vena cava. The suturing device **1100**′ can be configured such that when the second guidewire **1010** is advanced into the superior vena cava, the second guidewire **1010** directs the device **1100**′ toward the center of the septum segundum **6** as the device **1100**′ is advanced.
(224) With the suturing device **1100**′ positioned with the suture clasp arm **1031**B′ extending around the tip of the septum secundum **6**, as illustrated in FIG. **29**K, the suture clasp arm **1031**B′ holds a suture portion **52**B extending from opening **1042**′ in suture clasp **1033**B′ such that the suture portion **52**B is positioned on the opposite side of the septum secundum **6** relative to the PFO **8** and the suturing device **1100**′.
(225) Once the suture clasp arm **1031**B′ and suture portion **52**B have been properly positioned around the septum secundum **6**, the needle **1165**′ may be deployed from the distal end of the suturing device **1100**′ to penetrate the septum secundum **6** and engage the suture portion **52**B. As shown in FIG. **29**L, the tip of the needle **1165**′ is pulled proximally from a location distal the suture clasp arm **1031**B′ through the tissue of the septum secundum **6** towards deployed suture clasp arm **1031**B′ to engage the suture portion **52**B.
(226) Once the suture portion **52**B has been engaged, the needle **1165**′ and engaged suture portion **52**B are then retracted distally through the tissue of the septum secundum **6** and into the cavity **1083**′ of the suturing device **1100**′, as illustrated in FIG. **29**M. The suture clasp arm **1031**B′ may then be closed and thereafter the second guidewire **1010** retracted into the suturing device **1100**′. Alternatively, the second guidewire **1010** can be retracted into the suturing device **1100**′ before the suture clasp arm **1031**B′ is closed. Once the suture clasp arm **1031**B′ is closed, the suturing device **1100**′ may be withdrawn from the patient's heart and the PFO can be closed as described above.
(227) An embodiment of a device **2100** for applying a knot to a suture, comprising a handle **2110** and an elongate tubular member **2120**, is illustrated in FIGS. **30-32**, which is similar in some respects to the devices described in U.S. Patent Publication No. 2007-0010829 A1, published Jan. 11, 2007, but differs in some respects. The device **2100** can be used to apply a knot to a suture in any of the methods described above.
(228) The handle **2110** of the device **2100** comprises a housing **2130**, a button **2140**, and a knob **2150**. Referring to FIG. **31**, the housing **2130** comprises an aperture **2132** configured to allow longitudinal and rotational movement of the knob **2150**, a passage **2134** for movement of the button **2140** into and out of the housing **2130**, and a mounting recess **2136**.
(229) The mounting recess **2136** can accommodate, support, and limit them movement of a mounting hub **2160**. The mounting hub **2160** can be fixedly attached to the elongate member **2120**. An actuating rod **2180** can extend through the elongate member **2120** to a shaft **2152** that is connected to the knob **2150**. The shaft **2152** can be fixedly attached to the actuating rod **2180**.
(230) The button **2140** can comprise a pin **2142**. When the button **2140** is positioned within the passage **2134**, the pin **2142** can extends into a recess **2135** of the passage **2134**. The recess **2135** can be dimensioned such that rotation and translation of the pin **2142** and button **2140** within the passage **2134** is constrained such that the button **2140** is not forced out of the passage **2134** by a

spring **2144** that is positioned in the passage **2134** between the button **2140** and the housing **2130**. In some embodiments, the button **2140** can have a hook, or projection, **2146** to engage the shaft **2152**.
(231) The aperture **2132** in the housing **2130** is configured to allow a user to manipulate at least a portion of the knob **2150** and can extend along the longitudinal axis of the device **2100** into the passage **2134**. Within the aperture **2132**, the housing **2130** can have a neck or ring **2138** that slidably and rotatably supports the shaft **2152**. A spring **2137** biases the knob **2150** distally from the neck **2138**.
(232) In some embodiments, the aperture **2132** can have a narrow region **2133** and a wide region **2135**. The narrow region **2133** can be dimensioned to inhibit rotation of the shaft **2152** when a pin **2154**, which extends from the shaft **2152**, is located within the narrow region **2133**. As the shaft **2152** moves proximally within the aperture **2132**, the pin **2154** moves from the narrow region **2133**, in which rotation of the pin **2154** is limited, into the wide region **2135**, which is large enough to allow the pin **2154** to rotate with the shaft **2152**.
(233) The shaft **2152** can also include a recess or groove **2156** for operative engagement with the hook **2146** of the button **2140** to retain the shaft **2152** in a proximally retracted position relative to the handle **2130**. When the button **2140** is advanced into the housing **2130**, the hook **2146** releases from the recess **2156** of the shaft **2152** and the spring **2138** biases the shaft **2152** distally within the handle **2130**, which in turn advances the actuator rod **2180** distally.
(234) As the shaft **2152** advances distally, the pin **2154** moves within the aperture **2132**. Once the pin **2154** advances from the narrow region **2133** into the wide region **2135**, the knob **2150** can be rotated.
(235) FIG. **32** illustrates the distal end of the elongate tubular member **2120**, the actuator rod **2180**, a tip **2184**, and a sleeve **2190**. A suture (not shown) can extend through an opening **2122** in the elongate member **2120**, an opening **2192** in the sleeve **2190**, and out of an opening **2124** in the distal end of the elongate member **2120**.
(236) The sleeve **2190** can support a knot body (not shown) and a plug (not shown) of the type disclosed in U.S. Patent Publication No. 2007-0010829 A1 with the suture (not shown) extending between the knot body (not shown) and the plug (not shown). As the actuator rod **2180** advances, the plug is forced into the knot body by the tip **2184** to secure the suture (not shown) between the knot body and plug in the manner described in U.S. Patent Publication No. 2007-0010829 A1.
(237) After the suture has been secured between the knot body and the plug, the actuator rod **2180** can be rotated by the user by rotating the knob **2150**. While the actuator rod **2180** is in the advanced position, rotation of the rod **2180** can cause a cutting edge **2182** of the tip **2184** to rotate toward the opening **2192** in the sleeve **2190**. The opening **2192** can have a cutting edge **2194** that cooperates with the cutting edge **2182** of the actuator rod **2180** to cut the suture.
(238) FIGS. **33**, **34**A, **34**B and **35** illustrate one embodiment of a handle **2200** that can be used in conjunction with a suturing device. For example, the handle **2200** can be connected to the elongate tubular member **1020** and spreader assembly **1090**, shown in FIGS. **14**A and **15**A, or the elongate tubular member **1020**′ and spreader assembly **1090**′, shown in FIGS. **14**B and **15**B. While the handle **2200** will be described in reference to suturing devices, the handle **2200** could be used in conjunction with devices for other purposes.
(239) As discussed above, in some embodiments, bending of the elongate tubular member can affect the relative positions of the ends of the suture catch mechanisms and the spreader assembly. For example, if an elongate tubular member is bent then a distal end of a suture catch mechanism extending along the inside of the bend in the elongate member relative to the central axis of the elongate tubular member would be advanced relative to the spreader assembly. In such a circumstance, the suture catch mechanism may be advanced through a suture clasp farther than is desired, which may result in enlargement of a loop at an end of a suture that in turn inhibits the ability of the suture catch mechanism to retract the and of the suture.

(240) On the other hand, if an elongate tubular member is bent then a distal end of a suture catch mechanism extending along the outside of the bend in the elongate member relative to the central axis of the elongate tubular member would be retracted relative to the spreader assembly. In such a circumstance, the suture catch mechanism may not be advanced through a suture clasp far enough to engage a suture end portion held by the suture clasp.
(241) As discussed above, in some embodiments, the effects of bending of the elongate tubular member can be reduced or eliminated by using a suture catch mechanism that is sufficiently long, providing a stop mechanism, or both. Additionally or alternatively, the handle **2200** can reduce or eliminate the effects of bending of the elongate tubular member.
(242) The handle **2200** can have a housing **2201**, and one or more actuators **2210**, **2216**, **2240**, and **2260** that extend from the housing **2201**, as illustrated in FIGS. **33** and **34**A. One or more pulls **2210** and **2216** can be actuated to deploy and retract the suture catch mechanisms, such as, needles **1161**, **1165** of the suturing device. One or more switches **2240**, **2260** can be actuated to deploy and retract the suture claps arms **1031**A and **1031**B.
(243) With reference to FIG. **33**, the housing **2201** can be attached to the proximal end of the elongate tubular member **1020**. The housing **2201** can have an aperture **2202** providing a passageway between the handle **2200** and the multiple lumens of the elongate tubular member **1020**.
(244) In some embodiments, the handle **2200** can comprise one or more ports **2205** located distal to the housing **2201**. The ports **2205** can comprise lumens configured to provide access to one or more lumens of the elongate tubular member **1020** for one or more of sutures, guidewires, or die.
(245) Referring to FIG. **34**A, the elongate member **1020** is fixed to the housing **2201** in any suitable manner such as by welding, adhesion, or other bonding process to a collar or cylinder **2203**. The cylinder **2203** is engaged by the housing **2201**. The cylinder **2203** can be rectangular or any other shape. The cylinder **2203** is connected to the housing **2201** in a manner that prevents limits rotation of the cylinder **2203** with the member **1020** relative to the housing **2201**.
(246) The elongate member **1020** has a plurality of cutouts (not shown) that expose the actuator rods **1035** and the suture catch mechanisms **1161** and **1165** to permit connection to the actuators **2210**, **2216**, **2240**, and **2260**. Referring to FIG. **15**A, the proximal ends of the needles **1161** and **1165** extend through the lumens **1021** and **1023** of the elongate tubular member and terminate at a connection to pulls **2210** and **2216** on the suturing device handle **2200**.
(247) In one embodiment, the needles **1161**, **1165** are connected to shafts, **2270**, **2276**, respectively. The proximal ends of needles **1161**, **1165** can extend from the proximal end of the lumens **1021**, **1023**, respectively, in the elongate tubular member **1020** through opening **2202** into the handle **2200** (FIG. **33**). The needles **1161**, **1165** can be exposed by cutouts in the elongate tubular member **1020** for connection to the shafts, **2270**, **2276**, respectively. Referring to FIGS. **34**A and **34**B, the shafts **2270**, **2276** can each have a passage to permit the elongate member **1020** to pass through the shafts **2270**, **2276**. The needles **1161**, **1165** can be connected to the shafts **2270**, **2276** directly by a suitable bonding process, or indirectly, such as by connection to small tubes that are adhered to the needles **1161**, **1165**, for example, by epoxy, which are in turn connected to tabs **2280**, **2286**. The tabs **2280**, **2286** can be connected to the shafts **2270**, **2276**, respectively, at notches **2290**, **2296** in the shafts **2270**, **2276**.
(248) The shafts **2270**, **2276** can be moved by the pulls **2210**, **2216** in cooperation with springs **2228**, **2238**. The pulls **2210**, **2216** can extend from the housing through openings **2213** and **2217**, respectively, as shown in FIG. **33**. Referring to FIG. **34**A, the pulls **2210**, **2216** can each comprise an external portion **2220**, **2226** and an internal portion **2230**, **2236**, which can be integrally formed or can be connected by any bonding process. The external portions **2220**, **2226** can be engaged by a user to move the pulls **2210**, **2216**, respectively.
(249) The internal portions **2230**, **2236** of the pulls **2210**, **2216** can each have a passage **2204**, **2206**, a first end portion **2221**, **2231**, and a second end portion **2223**, **2233**. The passages **2204**,

2206 can be configured to allow the shafts **2270**, **2276** to move therein and the elongate tubular member **1020** to extend through the pulls **2210**, **2216**. The first end portions **2221**, **2231** and second end portions **2223**, **2233** can limit the movement of the shafts **2270**, **2276** within the passage.

(250) The first end portions **2221**, **2231** and second end portions **2223**, **2233** can be integrally formed with the internal portions **2230**, **2236**, or can be separate components. For example, in the embodiment of FIGS. **34**A and **34**B, the first end portions **2221**, **2231** are integrally formed with the internal portions and second end portions **2223**, **2233** are illustrated as separate components which can be mechanically connected, such as by threads, or can be connected by adhesive, or other bonding process to the internal portions.

(251) The springs **2228**, **2238** can be positioned between the second end portions **2223**, **2233** and the shafts **2270**, **2276**, thereby biasing the shafts **2270**, **2276** toward the first end portions **2221**, **2231**. In some embodiments, collars **2227**, **2237** can be positioned between the springs **2228**, **2238** and the shafts **2270**, **2276**. The collars **2227**, **2237** may slidably move over the elongate tubular member **1020** and provide an interface between the springs **2228**, **2238** and the shafts **2270**, **2276**. In some embodiments, a spring **2218** can be positioned between the pulls **2210** and **2216**. The spring **2218** can bias the needle **1161** proximally and the needle **1165** distally by biasing the pull **2210**, which is connected to the needle **1161**, from the pull **2216**, which is connected to the needle **1165**.

(252) In the embodiment illustrated in FIGS. **34**A and **34**B, a user can advance the needle **1161** by moving the pull **2210** to compress the spring **2218**. As the pull **2210** is advanced distally, the second end portion **2223** pushes against the spring **2228**, which in turn pushes the shaft **2270** distally along with the needle **1161**. However, distal advancement of the shaft **2270** is limited by the first end portion **2221**. Thus, a user is able to advance the needle **1161** at a controlled rate and is able to retract the needle **1161** by moving the pull **2210** proximally.

(253) On the other hand, should the needle **1161** encounter resistance as the pull **2210** is advanced, such as from a stop mechanism, described above, the shaft **2270** will move away from the first end portion **2221** toward the second end **2223**. Thus, the internal spring **2228** is compressed to avoid over-extension of the needle **1161**.

(254) Likewise, with continued reference to FIGS. **34**A and **34**B, a user can draw the needle **1165** proximally by moving the pull **2216** to compress the spring **2218**. As the pull **2216** is drawn proximally, the second end portion **2233** pushes against the spring **2238**, which in turn pushes the shaft **2276** proximally along with the needle **1165**. However, proximal movement of the shaft **2276** is limited by the first end portion **2231**. Thus, a user is able draw the needle **1165** proximally at a controlled rate and is able to retract the needle **1165** by moving the pull **2210** distally.

(255) Should the needle **1165** encounter resistance as the pull **2216** is advanced, such as from a stop mechanism, described above, the shaft **2276** will move away from the first end portion **2231** toward the second end **2233**. Thus, the internal spring **2238** is compressed to avoid over-extension of the needle **1165**.

(256) In some embodiments, the pulls **2210**, **2216** can comprise slots **2224**, **2234**, while the shafts **2270**, **2276** can comprise tabs **2225**, **2235** that have holes therein. Thus, a user may place a tool through the hole in the tab **2225**, **2235** to move the needle **1161**, **1165** directly, without the aid of the pulls **2210**, **2216** or the springs **2228**, **2238**.

(257) Referring to FIGS. **33**, **34**A, **34**B, the suture claps arms **1031**A and **1031**B can be deployed and retracted by movement of actuators **2240** and **2260**. The actuators **2240**, **2260** can be levers, switches, or rockers. In other embodiments the actuators **2240**, **2260** can have other configurations.

(258) As noted above, the elongate member **1020** can have a plurality of cutouts (not shown) that expose the actuator rods **1035** to permit connection to the actuators **2240**, **2260**. Referring to FIG. **15**A, the proximal ends of the actuator rods **1035**A, **1035**B extend through the lumens **1026**, **1024** of the elongate tubular member and terminate at connections to the pulls **2240** and **2260** on the suturing device handle **2200** (FIG. **33**).

(259) In one embodiment, the actuator rods **1035**A, **1035**B, which are connected to the suture claps arms **1031**A and **1031**B, are also connected to shafts **2245**, **2265** (FIGS. **34**A, **34**B), respectively. The proximal ends of the actuator rods **1035**A, **1035**B can extend from the proximal end of the lumens **1026**, **1024**, respectively, in the elongate tubular member **1020** through opening **2202** into the handle **2200**. The actuator rods **1035**A, **1035**B can be exposed by cutouts (not shown) in the elongate tubular member **1020** for connection to the shafts **2245**, **2265**, respectively. Referring to FIGS. **34**A and **34**B, the shafts **2245**, **2265** can each have a passage to permit the elongate member **1020** to pass through the shafts **2245**, **2265**. The actuator rods **1035**A, **1035**B can be connected to the shafts **2245**, **2265** directly by a suitable bonding process, or indirectly, such as by connection to small tubes that are adhered to the actuator rods **1035**A, **1035**B, for example, by epoxy, which are in turn connected to tabs **2246**, **2266**. The tabs **2246**, **2266** can be connected to the shafts **2245**, **2265**, respectively, at notches **2247**, **2267** in the shafts **2245**, **2265**. Thus, the actuator rods **1035**A, **1035**B move according to the movement of the shafts **2245**, **2265**.
(260) With continued reference to FIGS. **34**A and **34**B, the handle **2200** can comprise springs **2251**, **2253**, **2271**, **2273**, collars **2252**, **2254**, **2272**, **2274**, and followers **2255**, **2275**. The springs **2251**, **2271** can located proximal to the shafts **2245**, **2265** between the housing **2201** the shafts **2245**, **2265**. The distal springs **2253**, **2273** can be located distal to the shafts **2245**, **2265** between the housing **2201** and the shafts **2245**, **2265**.
(261) The collars **2252**, **2272** can be located proximal to the shafts **2245**, **2265**, between the proximal springs **2251**, **2271** and the shafts **2245**, **2265**. Thus, the proximal springs **2251**, **2271** tend to distally bias the proximal collars **2252**, **2272** against the shafts **2245**, **2265**.
(262) The collars **2254**, **2274** can be located distal to the shafts **2245**, **2265** between the distal springs **2253**, **2273** and the shafts **2245**, **2265**. Thus, the distal springs **2253**, **2273** tend to proximally bias the distal collars **2254**, **2274** against the shafts **2245**, **2265**.
(263) With reference to FIG. **34**A, the followers **2255**, **2275** can each comprise a passage **2258**, **2278**, shoulders **2259**, **2279**, and ramps **2257**, **2277** (FIG. **35**). The passages **2258**, **2278** can be configured to allow the shafts **2245**, **2265** to move through the passages **2258**, **2278**, and the elongate tubular member **1020** to extend through the followers **2255**, **2275**. The shoulders **2259**, **2279** and collars **2252**, **2254**, **2272**, **2274** can be configured such that the collars engage the shoulders to prevent the collars from passing completely through the followers **2255**, **2275**.
(264) Referring to FIG. **35**, the actuators **2240**, **2260** can each comprise cams **2241**, **2261** and pivot pins **2249**, **2269**. The pivot pins **2249**, **2269** cooperate with the housing **2201** for pivoting movement of the actuators **2240**, **2260**. The cams **2241**, **2261** and the ramps **2257**, **2277** of the followers **2255**, **2275** can be configured such that cams engage the ramps to move the followers as the actuators pivot relative to the housing.
(265) In use, movement of the actuators **2240**, **2260** allows the springs **2251**, **2253**, **2271**, **2273** to move shafts **2245**, **2265** proximally and distally. For example, with reference to FIG. **34**A, as the actuators **2240**, **2260** pivot counterclockwise about the pivot pins **2249**, **2269** (FIG. **34**B), the followers **2255**, **2275** are pushed proximally. As the followers **2255**, **2275** move proximally, the proximal collars **2252**, **2272** are likewise moved proximally to compress the proximal springs **2251**, **2271**. Proximal movement of the followers **2255**, **2275** also allows the distal springs **2253**, **2273** to proximally advance the distal collars **2254**, **2274**. In turn, the distal collars **2254**, **2274** proximally advance the shafts **2245**, **2265**. As described above, the suture clasp arms **1031**A, **1031**B move with the shafts **2245**, **2265**. Thus, the suture clasp arms **1031**A, **1031**B are moved to their extended positions.
(266) The distal springs **2253**, **2273** can be configured such that they will not proximally move the actuator rods **1035**A, **1035**B beyond the location where the suture clasp arms **1031**A, **1031**B are fully extended, even if the actuators **2240**, **2260** move the followers **2255**, **2275** far enough to allow further decompression of the distal springs. For example, the force exerted by the distal springs can be insufficient to extend the suture clasp arms **1031**A, **1031**B beyond their fully-extended

positions. On the contrary, the springs **2253**, **2273** can have sufficient length and apply sufficient force to fully extend the suture clasp arms **1031**A, **1031**B, even if the elongate member **1020** is bent.

(267) With continued reference to FIG. **34**A, as the actuators **2240**, **2260** pivot clockwise about the pivot pins **2249**, **2269** (FIG. **34**B), the followers **2255**, **2275** are pushed distally. As the followers **2255**, **2275** move distally, the distal collars **2254**, **2274** are likewise moved distally to compress the distal springs **2253**, **2273**. Distal movement of the followers **2255**, **2275** also allows the proximal springs **2251**, **2271** to distally advance the proximal collars **2252**, **2272**. In turn, the proximal collars **2252**, **2272** distally advance the shafts **2245**, **2265**. As described above, the suture clasp arms **1031**A, **1031**B move with the shafts **2245**, **2265**. Thus, the suture clasp arms **1031**A, **1031**B are moved to their retracted positions.

(268) The proximal springs **2251**, **2271** can be configured such that they will not distally advance actuator rods **1035**A, **1035**B beyond the location where the suture clasp arms **1031**A, **1031**B are fully retracted, even if the actuators **2240**, **2260** move the followers **2255**, **2275** far enough to allow further decompression of the proximal springs. For example, the force exerted by the proximal springs can be insufficient to damage the actuator rods **1035**A, **1035**B or the suture clasp arms **1031**A, **1031**B once the suture clasp arms have reached their fully-retracted positions. On the contrary, the proximal springs **2251**, **2271** can have sufficient length and apply sufficient force to fully retract the suture clasp arms **1031**A, **1031**B, even if the elongate member **1020** is bent.

(269) FIG. **36** illustrates one embodiment of a system for suturing an opening in a vessel wall or other biological tissue, or for performing other procedures as described above. The system **3000** can comprise a first suturing device **3100** and a second suturing device **4100**. While the system and suturing devices **3100**, **4100** will be described in reference to suturing an opening in the heart wall, such as a patent foramen ovale (PFO), the first suturing device **3100** and the second suturing device **4100**, either alone or in combination, could be used, like the devices **100**, **1100**, **1100′**, to close other openings in the heart wall, such as a patent ductus arteriosus (PDA) or an atrial septal defect (ASD), other openings in bodily tissue, or the like. The device **1100** could also be used to suture adjacent biological structures or any other time it may be desired to apply a suture to a biological structure.

(270) The first suturing device **3100** and the second suturing device **4100** can be similar to the suturing devices **1100**, **1100′** in some respects. Accordingly, components of the first suturing device **3100** and component of the second suturing device **41000** that are similar to those of the suturing devices **1100**, **1100′** are indicated by similar reference numerals 31XX and 41XX, respectively, rather than 11XX. For example, the suturing devices **3100**, **4100** can each comprise an elongate tubular member **3020**, **4020** having a spreader assembly **3090**, **4090** connected to the distal end of the elongate tubular member **3020**, **4020**. The elongate tubular members **3020**, **4020** can be similar to the elongate tubular members **1020**, **1020′**. Likewise, the spreader assemblies **3090**, **4090** can be similar to the spreader assemblies **1090**, **1090′**. A handle **3200**, **4200** can be provided at the proximal end of each tubular member **3020**, **4020**. The handles **3200**, **4200** can be similar to the handle **1200** or the handle **2200**.

(271) The first suturing device **3100** and the second suturing device **4100** can differ from the suturing devices **1100**, **1100′** in some respects. For example, in contrast to the devices **1100**, **1100′**, the devices **3100**, **4100** can comprise a single suture clasp arm and a single suture catch mechanism. Accordingly, the elongate tubular members **3020**, **4020**, spreader assemblies **3090**, **4090**, and handles **3200**, **4200** can be configured to operate the single suture clasp arm and the single suture catch mechanism, rather than plural suture clasp arms and plural suture catch mechanisms, by omission of those components associated with more than a single suture clasp arm and a single suture catch mechanism.

(272) The spreader assemblies **3090**, **4090** of the devices **3100**, **4100** can be shorter than the spreader assemblies **1090**, **1090′** because the spreader assemblies **3090**, **4090** may comprise fewer

components the spreader assemblies **1090**, **1090′**. Therefore, the spreader assemblies **3090**, **4090** may extend into left atrium less than the spreader assemblies **1090**, **1090′** when positioned in a PFO.

(273) The first suturing device **3100** can comprise a suture clasp arm **3031**A and a suture catch mechanism **3161** (FIG. **39**C) operated by a handle **3200** having a pull **3210** and an actuator **3240**. Referring to FIG. **37**, the suture clasp arm **3031**A can comprise a deflecting plate **3099** that is integrally formed with the suture clasp arm **3031**A.

(274) The second suturing device **4100** can comprise a suture clasp arm **4031**B and a suture catch mechanism **4165** (FIG. **39**G) operated by a handle **4200** having a pull **4216** and an actuator **4260**. The elongate tubular member **4020** and the spreader assembly **4090** can be configured for use of two guidewires similar to the elongate tubular member **1020′** and the spreader assembly **1090′**. Thus, the spreader assembly **4090** can comprise an opening **4043**, shown in FIG. **38**) through which a second guidewire can extend.

(275) The suture clasp arm **4031**B can comprise a guide section **4032**, as shown in FIG. **38**. The guide section **4032** can assist in maintaining a desired rotational orientation between the second guidewire and the spreader assembly **4090**. The guide section **4032** can thus assist a user in directing the second guidewire to a desired location, such as the superior vena cava, as the second guidewire is extended from the device **4100**. Additionally or alternatively, the guide section **4032** can assist a user in positioning the device **4100** in a desired located, such as substantially centered relative to the septum secundum **6** of the PFO.

(276) The operation of the system **3000** comprising the first suturing device **3100** and the second suturing device **4100**, described above, is illustrated according to one embodiment in FIGS. **39**A-**39**L in conjunction with a procedure for closing a PFO. As shown in FIG. **39**A, the distal end of a suturing device **3100** is advanced through the vasculature and positioned in the tunnel **8** of the PFO between the septum primum **7** and the septum secundum **6**. The suturing device **3100** may be advanced over a guidewire **10** or alternatively delivered through a catheter introducer sheath **1011** using techniques which are known in the art.

(277) The suturing device **3100** is initially positioned such that the spreader assembly **3090** is near the tip of the secundum primum **7** and the suture clasp deployment arm **3031**A is permitted to extend from the spreader assembly **3090**. The suture clasp arm **3031**A may then deployed from the spreader assembly **3090** and then the device **3100** is retracted until the suture clasp arm **3031**A extends around the tip of the secundum primum **7**, as shown in FIG. **39**B, and gathers the tissue of the septum primum **7** between the arm **3031**A and the spreader **3030**.

(278) Once the suture clasp arm **3031**A has been properly positioned around the septum primum **7**, needle **3161** may be deployed from the suturing device **3100** to penetrate the septum primum **7** and engage the suture clasp arm. The needle **3161** is advanced through a passageway in the suturing device and deflected by needle guide **3060**A along an angle that intersects the deployed suture clasp arm **3031**A as it exits the suturing device **3100**. The needle engages the suture clasp arm **3031**A, as shown in FIG. **39**C, to engage the suture portion **3052**A.

(279) As shown in FIG. **39**D, once the suture portion **3052**A has been engaged, the needle **3161** and engaged suture portion **3052**A are then retracted through the tissue of the septum primum **7** into the elongate member **3020** of the suturing device **3100**. The device **3100** may be advanced slightly so that the suture clasp arm **3031**A can be closed without pinching the septum primum **7**. The first suturing device **3100** may then be withdrawn from the vasculature over the guidewire **10**.

(280) The second suturing device **4100** may then be advanced through the venous access into the tunnel **8** of the PFO between the septum primum **7** and the septum secundum **6**, as shown in FIG. **39**E. The second guidewire **1010** can be advanced through the opening **4043** (FIG. **38**) into the superior vena cava (FIG. **1**). The second guidewire **1010** may be preloaded in the device **4100** before introduction of the device **4100** into the body such that a distal end of the second guidewire is near the opening **4043**.

(281) The suture clasp arm **4031**B can then be extended and the device **4100** can be advanced such that the suture clasp arm **4031**B extends around the tip of the septum secundum **6**, as shown in FIG. **39**F, and gathers the tissue of the septum secundum **6** between the arm **4031**B and the spreader **4030**. Alternatively, the suture clasp arm **4031**B can be extended from the spreader assembly **4090** before the second guidewire **1010** is advanced into the superior vena cava. The suturing device **4100** can be configured such that when the second guidewire **1010** is advanced into the superior vena cava, the second guidewire **1010** directs the device **4100** toward the center of the septum segundum **6** as the device **4100** is advanced.
(282) Once the suture clasp arm **4031**B and suture portion **4052**B have been properly positioned around the septum secundum **6**, the needle **4165** may be deployed from the distal end of the suturing device **4100** to penetrate the septum secundum **6** and engage the suture portion **4052**B. As shown in FIG. **39**G, the tip of the needle **4165** is pulled proximally from a location distal the suture clasp arm **4031**B through the tissue of the septum secundum **6** towards deployed suture clasp arm **4031**B to engage the suture portion **4052**B, as shown in FIG. **39**G.
(283) The suture portion **4052**B has been engaged, the needle **4165** and engaged suture portion **4052**B are then retracted distally through the tissue of the septum secundum **6** and into the suturing device **4100**, as illustrated in FIG. **39**H. The suture clasp arm **4031**B may then be closed and thereafter the second guidewire **1010** retracted into the suturing device **4100**. Alternatively, the second guidewire **1010** can be retracted into the suturing device **4100** before the suture clasp arm **4031**B is closed. Once the suture clasp arm **4031**B is closed, the suturing device **4100** may be withdrawn from the patient's heart.
(284) As shown in FIG. **39**I, the suture portion **3052**A has been positioned through the septum primum **7** while suture portion **4052**B has been positioned through the septum secundum. After the suturing device **4100** has been withdrawn, the suture portions **3052**A, **3052**B, **4052**A and **4052**B will extend proximally from the PFO. The suture portions **3052**B and **4052**A can then be secured together, as illustrated in FIG. **39**J, by tying a knot according to any known method or by applying a knot by any method or device that is described or referenced herein. The suture portions **3052**B and **4052**A can be secured together exterior to the body or within the body. Any excess portion of sutures **3050** and **4050** can be trimmed.
(285) The suture portions **3052**A and **4052**B can and can then be pulled to draw the septum secundum **6** and septum primum **7** towards one another to close the PFO, as described above. As the sutures **3050**, **4050** are pulled tight, the sutures **3050**, **4050** preferably cause the septum secundum **6** and septum primum **7** to turns or folds so that the tip of the septum primum **7** extends in the opposite direction compared to the tip of the septum secundum **6**, as shown in FIG. **39**K. The knot can be positioned between the septum primum **7** and the septum secundum **6**, as illustrated in FIG. **39**K. Such placement of the knot may agitate the tissue and promote healing between the septum primum **7** and the septum secundum **6**. In one embodiment, the knot is positioned between the septum primum **7** and the septum secundum **6**, as illustrated in FIG. **39**K, by pulling the suture portion **4052**B until the knot is pulled against the septum secundum **6** then pulling the suture portion **3052**A to those the PFO.
(286) An alternative method of operation of the second suturing device **4100** is illustrated according to one embodiment in FIGS. **40**A-**40**E. After operating the first suturing device **3100** according to the above description of FIGS. **39**A-D and withdrawn, the second suturing device **4100** may then be advanced through the venous access into the right atrium **2** of the heart adjacent to the PFO, as shown in FIG. **40**A. The first guidewire **10** can be withdrawn from the PFO and advanced into the superior vena cava (FIG. **1**) before or after introduction of the second suturing device **4100**. Additionally or alternatively, a portion of the device **4100** can be advanced into the superior vena cava.
(287) The second guidewire **1010** can be advanced from the opening **4043** (FIG. **38**) through the PFO between the septum primum **7** and the septum secundum **6**. The suture clasp arm **4031**B can

then be extended and the device **4100** can be advanced such that the suture clasp arm **4031**B extends around the tip of the septum secundum **6**, as shown in FIG. **40**B, and gathers the tissue of the septum secundum **6** between the arm **4031**B and the spreader **4030**. Alternatively, the suture clasp arm **4031**B can be extended from the spreader assembly **4090** before the second guidewire **1010** is advanced into the PFO. The suturing device **4100** can be configured such that when the first guidewire **10** is advanced into the superior vena cava and the second guidewire **1010** is positioned through the PFO, the first guidewire **10** and the second guidewire **1010** direct the device **4100** toward the center of the septum secundum **6** as the device **4100** is advanced.
(288) Once the suture clasp arm **4031**B and suture portion **4052**B have been properly positioned around the septum secundum **6**, the needle **4165** may be deployed from the distal end of the suturing device **4100** to penetrate the septum secundum **6** and engage the suture portion **4052**B. As shown in FIG. **40**C, the tip of the needle **4165** is pulled proximally from a location distal the suture clasp arm **4031**B through the tissue of the septum secundum **6** towards deployed suture clasp arm **4031**B to engage the suture portion **4052**B.
(289) As shown in FIG. **40**D, once the suture portion **4052**B has been engaged, the needle **4165** and engaged suture portion **4052**B are then retracted distally through the tissue of the septum secundum **6** and into the suturing device **4100**. The device **4100** can then be retracted slightly to permit the suture clasp arm **4031**B to closed. Thereafter, the second guidewire **1010** can be retracted into the suturing device **4100**. Alternatively, the second guidewire **1010** can be retracted into the suturing device **4100** before the suture clasp arm **4031**B is closed. Once the suture clasp arm **4031**B is closed, the suturing device **4100** may be withdrawn from the patient's heart.
(290) As shown in FIG. **40**E, the suture portion **3052**A has been positioned through the septum primum **7** while suture portion **4052**B has been positioned through the septum secundum. After the suturing device **4100** has been withdrawn, the sutures **3050** and **4050** can be used to draw the septum secundum **6** and septum primum **7** together to close the PFO, as described above.
(291) Although the operation of the devices **3100** and **4100** has been described with reference to two sutures **3050**, **4050**, the devices **3100**, **4100** can be used in some embodiments to place a single suture through both the septum primum **7** and the septum secundum **6**, or to place multiple sutures through each of the septum primum **7** and the septum secundum **6**. In some embodiments, plural devices **3100**, plural devices **4100**, or both can be used to place multiple sutures through one or both of the septum primum **7**, the septum secundum **6**, or other biological tissue, biological structure, prosthetic, or synthetic material or implantable device in the body. For example, plural devices may be used to suture a prosthetic heart valve to the heart or to affix a balloon, umbrella, or other device that is not properly positioned to the surrounding tissue.
(292) Although the foregoing description of the preferred embodiments of the present invention has shown, described and pointed out the fundamental novel features of the invention, it will be understood that various omissions, substitutions, and changes in the form of the detail of the apparatus as illustrated as well as the uses thereof, may be made by those skilled in the art, without departing from the spirit of the invention. For example, while the suturing device is described with respect to closing a patent foramen ovale in a patient's heart, it is further envisioned that the suturing device could be used to close or reduce a variety of other tissue openings, lumens, hollow organs or natural or surgically created passageways in the body. The suturing device may have any suitable number of arms, such as two or four or more, and any given arm may have one or more suture clasps or openings.

Claims

1. A suturing apparatus for suturing tissue, comprising: a first elongate body having a proximal end and a distal end and defining a first longitudinal axis; a first spreader assembly coupled to and extending from the distal end of the first elongate body such that the first spreader assembly is

coaxial with the first elongate body along the first longitudinal axis, the first spreader assembly comprising: a first suture clasp arm configured to hold a first suture end portion, the first suture clasp arm being pivotably moveable between a retracted position and an extended position relative to the first longitudinal axis; and a first suture catch mechanism slidably housed in said first elongate body, the first suture catch mechanism being movable in a first direction through a first portion of the tissue to engage a distal end of the first suture catch mechanism with the first suture end portion held by the first suture clasp arm when the first suture clasp arm is in the extended position prior to engagement with the first suture catch mechanism; and a second elongate body having a proximal end and a distal end, and defining a second longitudinal axis, the second elongate body is discrete from the first elongate body, the second elongate body comprising: a second suture clasp arm adapted to hold a second suture end portion, the second suture clasp arm being pivotably moveable between a retracted position and an extended position relative to the second longitudinal axis; and a second suture catch mechanism slidably housed in said second elongate body, the second suture catch mechanism being movable in a second direction through a second portion of the tissue to engage a distal end of the second suture catch mechanism with the second suture end portion held by the second suture clasp arm when the second suture clasp arm is in the extended position prior to engagement with the second suture catch mechanism.
2. The apparatus of claim 1, wherein the tissue is a patent foramen ovale.
3. The apparatus of claim 1, wherein the first elongate body further comprises a first guide configured to deflect the first suture catch mechanism when the first suture catch mechanism moves in the first direction.
4. The apparatus of claim 3, wherein the second elongate body further comprises a second guide configured to deflect the second suture catch mechanism when the second suture catch mechanism moves in the second direction.
5. The apparatus of claim 1, wherein the first direction is opposite to the second direction.
6. The apparatus of claim 1, wherein the first direction is a proximal-to-distal direction.
7. The apparatus of claim 6, wherein the second direction is a distal-to-proximal direction.
8. The apparatus of claim 1, wherein the second direction is a distal-to-proximal direction.
9. The apparatus of claim 1, wherein at least one of the first elongate body or the second elongate body is configured to be advanced over a guidewire.
10. The apparatus of claim 1, wherein the first suture clasp arm and the second suture clasp arm are separately moveable between the respective retracted positions and the extended positions.
11. The apparatus of claim 1, wherein: the first suture clasp arm in its extended position forms an acute angle with the longitudinal axis of the first elongate body, and extends in a proximal direction, and the second suture clasp arm in its extended position forms an acute angle with the longitudinal axis of the second elongate body, and extends in a distal direction.
12. The apparatus of claim 1, wherein the first elongate body is flexible.
13. The apparatus of claim 12, wherein the second elongate body is flexible.
14. The apparatus of claim 1, wherein the second elongate body comprises a second spreader assembly coupled to and extending from the distal end of the second elongate body such that the second spreader assembly is coaxial with the second elongate body along the second longitudinal axis, the second spreader assembly comprising the second suture clasp arm and the second suture catch mechanism.
15. The apparatus of claim 1, wherein the first suture clasp arm is the only suture clasp arm of the first elongate body.
16. The apparatus of claim 15, wherein the second suture clasp arm is the only suture clasp arm of the second elongate body.
17. The apparatus of claim 1, wherein the second suture clasp arm is the only suture clasp arm of the second elongate body.